

GS8 D2

Assembly guide

Digital Super 8 cine camera, D2 fork (cad/g8-d2-v1)

GS8 D2 r7-20261009

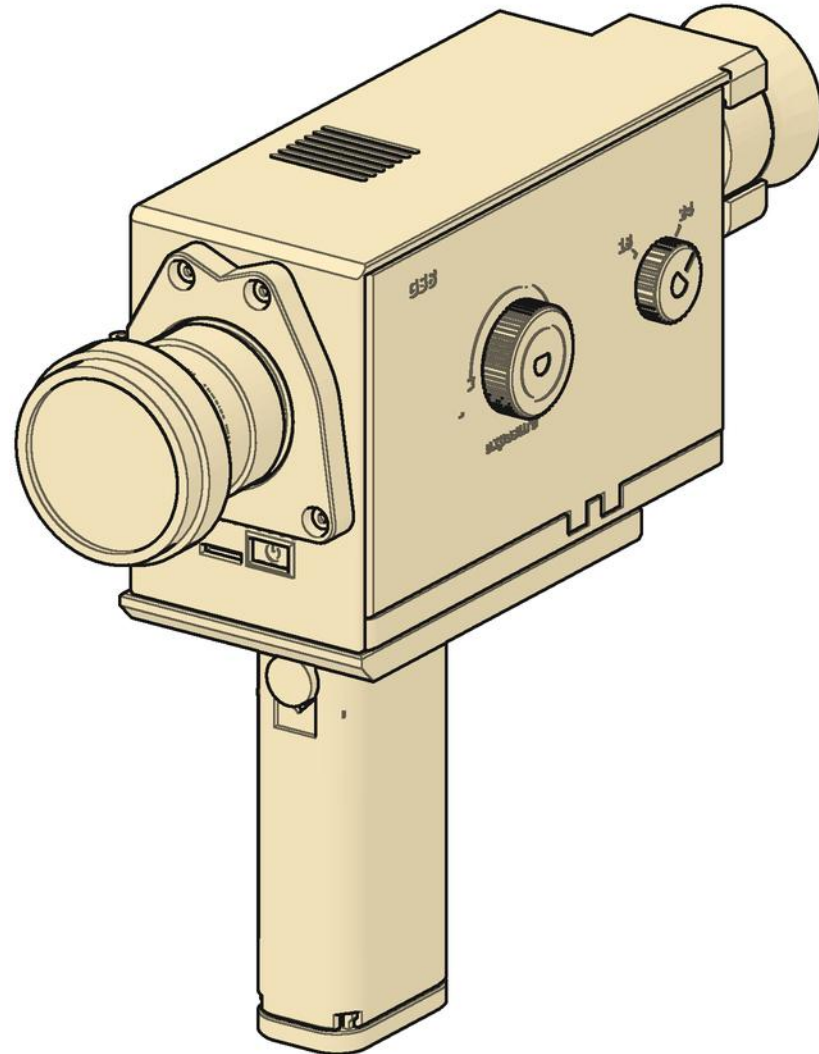
Guide built 2026-10-09 06:19

DRAFT. Nobody has assembled a D2 yet. The steps follow ASSEMBLY.md (r7) and WIRING.md s7 and are checked only by computed insertion sweeps and the straight-driver audit.

Read pages 0.1 to 0.7 before you start: rules, tools, hardware bags, parts, gates, contents.

Each step page has the picture on the left (new parts highlighted, arrows show the motion) and the written instructions on the right: YOU NEED, DO THIS, CAUTION, CHECK.

Directions in the text: forward = toward the lens, left = the panel side, as seen from behind the camera looking through the eyepiece.



0.1

Read first

Rules for the whole build (ASSEMBLY s1)

Power last

The battery pack goes in only at step 10, after the body is closed. Every lead in steps 1-9 is made with the pack out of the camera. Never plug or unplug an internal lead with the pack connected.

Straight driver only

Every screw is driven with a straight PH1 driver along its own axis. No L-keys, hex keys, angled or ball-end drivers. One bench exception: the camera tripod block screws at B0, if they turn out to be hex.

Torque

PT screws (plastic, thread-forming): 0.35-0.5 N m, stop at head contact; no threadlocker; do not re-torque. Lens collar M3: s_c1, s_c2, s_c3 0.15 N m, s_c4 0.2 N m, always on the torque screwdriver (tool 13).

Lens rule

Never thread the lens into an unsupported camera. Nobody holds the camera: one fingertip pushes the cover centre forward (push only), and the hood tab catch holds the camera through its metal lock tab while the lens turns. Never grip, pinch or turn the cover or the circuit board. Check both paint marks at every lens change.

ESD

Grounded mat and wrist strap for steps 1, 4, 6, 7 and bench step B0 (bare boards, OLED flex, camera).

Re-assembly

At most 5 drives per plastic boss. Tally every drive on the build sheet; after 5, use the M3 insert fallback (FASTENER-POLICY D-E).

Not yet built

Nobody has assembled a D2. These steps are checked by computed insertion sweeps and the straight-driver audit only. Stop and report if a part does not go where the picture shows.

0.2 Tools

14 tools (ASSEMBLY s1.1); the numbers are used on every step page

(1) PH1 screwdriver, straight (shank dia 6.5 or less, blade 40 mm or longer, handle dia 28-30)

BOM D2-60

(2) Temperature-controlled soldering iron + solder, flux, wick; M3 heat-set tip

BOM D2-63, D2-47, D2-74

(3) Wire stripper + flush cutter

BOM D2-68

(4) Heat gun (adhesive heat-shrink)

BOM D2-68

(5) Multimeter (continuity, diode test, no-short checks)

BOM D2-64

(6) 12.7 mm (1/2 in) socket or spanner (18/24 switch nut)

BOM D2-61

(7) Junior hacksaw + small flat file + caliper with depth rod + soft-jaw vise (cut the switch shaft off the panel)

BOM D2-69

(8) Fine tweezers, 120 mm

BOM D2-62

(9) 2 hood release pins, dia 1.5 steel, 16 mm or longer (service only)

BOM D2-71

(10) Flat bar or steel rule (press the tripod nut; upright HDMI push stick at 6c)

BOM -

(11) Paint pen (engraving fill, adapter mark)

BOM D2-44

(12) ESD mat + wrist strap

BOM D2-70

(13) Adjustable torque screwdriver 0.1-0.6 N m, straight, PH1 bit

BOM D2-77

(14) Collar centring gauge, printed, one per lens (stl/tools/collar_gauge.stl)

BOM printed

(15) Smooth-jaw long-nose pliers, jaws 35 mm or longer, tips 2.0 mm thick or less (EVF HDMI unplug, service)

BOM -

(16) Printer paper, 2 strips (about 0.2 mm together: knob stop shim)

BOM -

(17) Inspection mirror, head 15 mm or less (optional: see the EVF HDMI seat)

BOM -

0.3

Hardware bags

Sort the loose hardware into these bags before you start

Bag A: 7 x PT 3.0 x 12 thread-forming screw for plastics, pan head, PH1 (EJOT PT K30x12 class)

Used at: s_j (step 3), s_k1, s_k2 (step 4), s_b1, s_b2, s_r1, s_r2 (step 8c)

Bag B: 3 x M3 x 10 PH1 machine screw + flat washer

Used at: s_c1, s_c2, s_c3 (step 7a)

Bag C: 1 x M3 x 16 PH1 machine screw + flat washer

Used at: s_c4 (step 8a)

Bag D: 3 x M3 brass heat-set insert, short (D2-35)

Used at: tub front wall TL, TR, LL (B1)

Bag E: 1 x M3 brass heat-set insert, L 5.7 (D2-36)

Used at: lens collar lower lug (B1)

Bag F: 1 x X1203 kit: 4 standoffs + 8 M2.5 screws

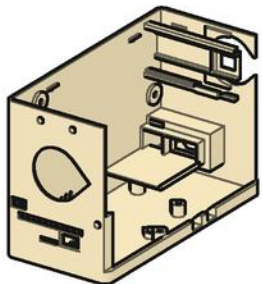
Used at: stack (step 1); if the kit screws are hex, M2.5 x 5 ISO 7045 PH1

Bag G: 1 x 1/4-20 UNC hex nut, steel

Used at: tripod nut (step 3a)

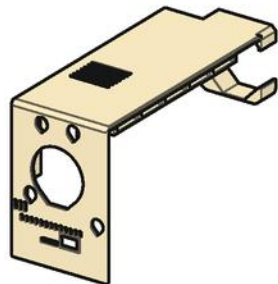
0.4 Printed parts

Check every part against this list before you start (not to scale)



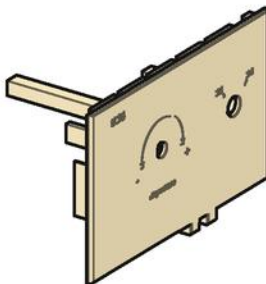
Tub (body)

ASA, satin silver



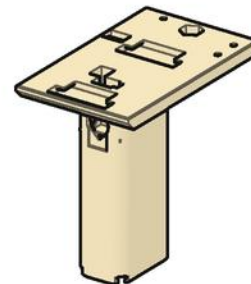
Hood

ASA, black



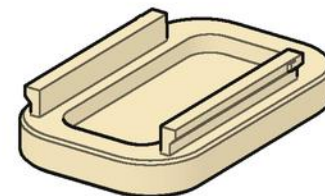
Panel

ASA, satin silver



Base + grip

ASA, black



Battery cap

ASA, black



Power plunger

ASA, black



Exposure knob

ASA, black



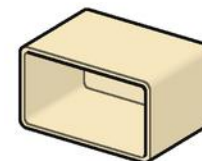
18/24 knob

ASA, black



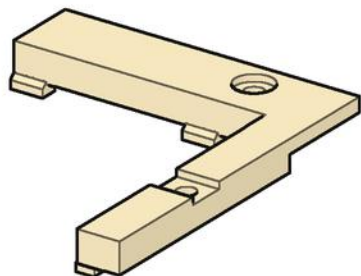
Eyecup

TPU 95A, black



USB stick sleeve

ASA, satin silver



Pi keeper

ASA, black

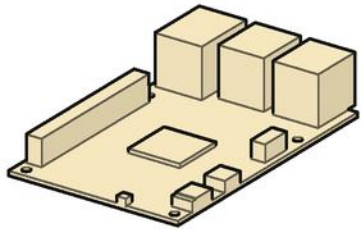


Lens collar

ASA (PC if G-W11 shows > 50 C at t

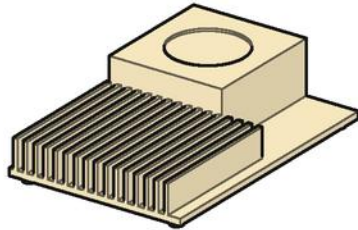
0.5 Purchased parts

Check every part against this list before you start (not to scale)



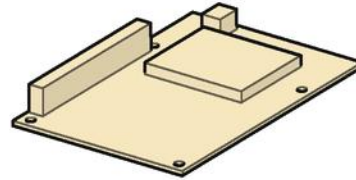
Raspberry Pi 5

SC1111/SC1112



Raspberry Pi Active Cooler

SC1148



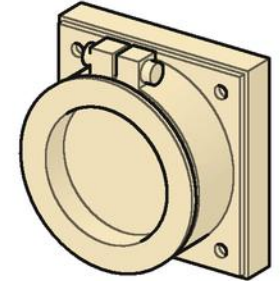
Geekworm X1203 UPS

X1203



X1203 kit standoffs + screws

kit



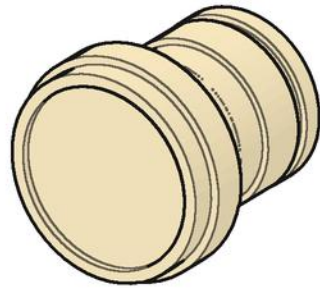
Raspberry Pi Global Shutter Camera

SC0926 class



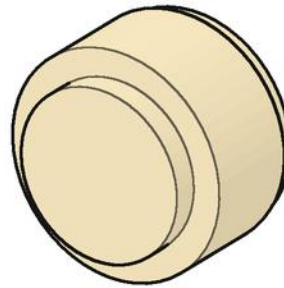
C-CS adapter ring 5 mm

kit



Kowa LM6HC 6 mm f/1.8 1in C-mount

see LENSES



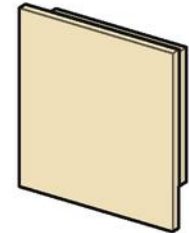
**Display Components
OPE039-16X glass eyepiece**

OPE039-16X



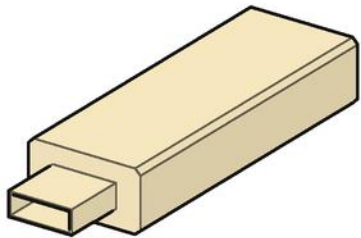
**Hicenda HMX039-V1
micro-OLED + 50 mm flex**

HMX039 kit



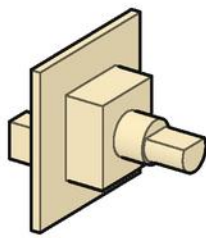
**Hicenda HDMI driver board
Rev I**

HMX039 kit



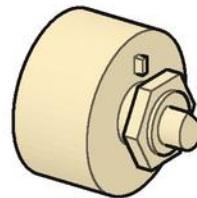
**SanDisk Extreme PRO USB 3.2
SSD stick**

SDCZ880



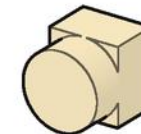
**Adafruit 5880 I2C QT rotary
encoder**

5880



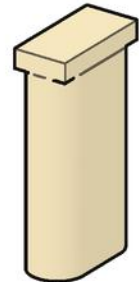
**mini 2-position rotary switch
+ nut + washer**

Lorlin CK1049 class



pre-wired momentary, red cap

Pi Hut Squid



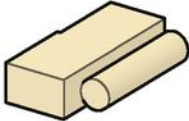
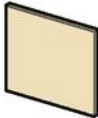
**1S2P 18650 pack, BMS, XT30
male lead, pull ribbon**

class

0.5

Purchased parts (continued)

Check every part against this list before you start (not to scale)

 XT30 pair XT30U; fuse Littelfuse 0251015.MXL	 1/4-20 UNC hex nut, steel ISO 4032 class	 12 mm hand strap with buckle class	 microSD card consumable class	 closed-cell foam 16 x 14 x 1.0 consumable
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0.6 Gates before assembly

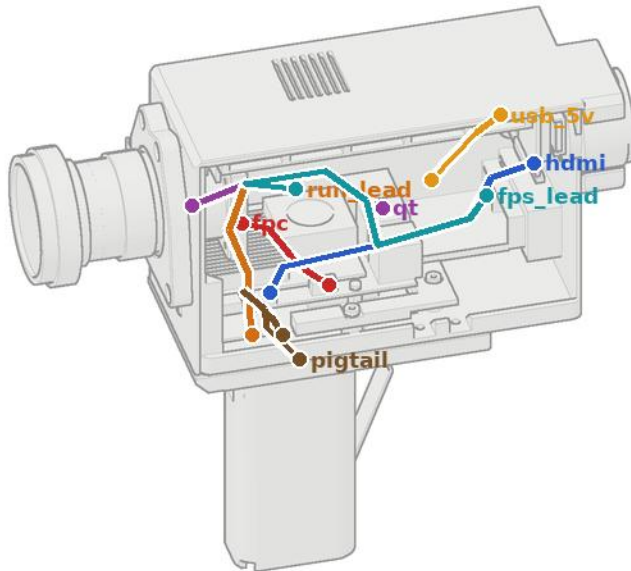
Do not start step 1 until these are recorded (ASSEMBLY s1)

- ☐ WIRING G-W1 to G-W7: pack, X1203 identity, Pi on the bench supply, stack with a pack supply (G-W4) and under load (G-W5), EVF feed (G-W6), HDMI pin 19 (G-W7).
- ☐ G-W13 only if EVF-feed option C is adopted (WIRING s4.8).
- ☐ SPEC G-CAM-1 and G-LENS: the measured camera and lens (MEASURED-PARTS MP-CAM with its lens line) before any lens collar, tub, hood or panel is printed. r7: G-CAM-1 also records the lock-screw head side and size, which set the hood tab catch.
- ☐ Bench (r7): G-W2 sets the pigtail cut length (pads-to-W, page 1a; more than 25 mm pads-to-W: stop, do not cut); G-MP-PACK measures the rigid solder-cup and heat-shrink length at each XT30 end (0.8 mm allowed) and adds a 10-cycle check of the folded pigtail above the pack (page 10a).
- ☐ Collar coupons G-COL-1, the print coupons (PRINT-GUIDE s6) and the measured-part records MP-PACK, MP-X1203, MP-CAM, MP-HDMI and the MP-EVF bench assembly.
- ☐ Open design item M1 (collar clamp spring and its thermal term) is to be fixed before the lens collar is printed.
- ☐ Before any claim that the powered camera works: WIRING G-W12 (full-workload power test).
- ☐ First print (r7): G-EVF-3, five in-body EVF HDMI mate/unmate cycles through the rail gap (page 6c, tools 10 and 15).
- ☐ First print (r7): G-QT-1 and G-HDR-1 (MEASURED-PARTS MP-QT): QT lead dry fit, 5 plug cycles with the panel 60 mm off (page 8b), 8 header housings level with a visible gap to the cooler and the wall (page 4d).

Wiring overview

Every cable, its route (keep-out chain) and the steps where it is plugged (WIRING s6-s7). Panel removed; routes drawn through the parts.

A from the left, above (panel off)



B from above



CABLES

- fpc (200 mm; plugged at step 4, 7)**
Pi 5 CAM/DISP 1 (port edge) -> GS camera (lower rear)
- hdmi (200 mm; plugged at step 4, 6)**
Pi 5 HDMI0 -> EVF board lower edge
- usb_5v (200 mm; plugged at step 4, 6)**
Pi 5 upper USB 2 port -> EVF board 5 V
- oled_flex (50 mm; plugged at step 6)**
HMX039 -> EVF board ZIF
- qt (250 mm; plugged at step 4, 8)**
encoder -> GPIO 1/3/5/6
- run_lead (250 mm; plugged at step 3, 4)**
run button -> GPIO26 + GND (pins 37/39)
- fps_lead (200 mm; plugged at step 1, 4, 8)**
18/24 switch -> GPIO13 + GND (pins 33/34)
- pigtail (200 mm; plugged at step 1, 4)**
X1203 battery pads -> XT30 junction in the grip
- fan (60 mm; plugged at step 1)**
Active Cooler -> Pi 5 FAN header
- pack_lead (60 mm; plugged at step 10)**
1S2P BMS -> XT30 male

Polarity and pin rules: header pins on page 4d, EVF lead on 1c, fuse on 1a, XT30 on 10b. Every lead is made with the pack out (power last).

0.7 Contents

B = bench; 1-10 = the build steps of layout.STEPS; letters are sub-steps

B0 Camera and lens on the bench

Bench, before any body part is printed (MP-CAM, G-CAM-1, G-LENS)

B1 Heat-set inserts

Bench, tub empty, any time before step 3 (collar insert: before step 7)

1a Fuse the battery pigtail

Step 1, bench wiring (WIRING s7)

1b Pigtail to the X1203 battery pads

Step 1, bench wiring

1c EVF 5 V lead with diode

Step 1, bench wiring

1d EVF board pigtail

Step 1, bench wiring

1e 18/24 switch pigtail

Step 1, bench wiring

1f Encoder address, QT lead splice

Step 1, bench wiring

1g Bench stack: X1203 + Pi 5 + cooler

Step 1, bench sub-assembly (joins the body at step 4)

2 Bench: panel

Step 2, bench sub-assembly (joins the body at step 8)

3a Base: tripod nut, run button, strap

Step 3

3b Tub onto the base, lock screw s_j

Step 3

4a Lead routing, then the Pi stack

Step 4 (ESD strap; microSD out)

4b Pi keeper, screws s_k1 and s_k2

Step 4

4c Plug the Pi end of HDMI, FPC and EVF 5 V lead

Step 4 (top open)

4d Header leads

Step 4 (top open, header in sight)

5a Plunger and hood

Step 5

5b microSD through the front slot

Step 5

6a Eyepiece

Step 6

6b Mate the EVF leads outside the body

Step 6 (ESD mat; pack out)

6c Slide the OLED and board in together

Step 6

7a Lens collar, centred

Step 7 (B0 done; camera still out)

7b Camera and adapter into the cage

Step 7 (ESD strap)

8a Lens through the collar, s_c4

Step 8 (panel still off)

8b Panel leads, then the panel

Step 8

8c Four panel screws

Step 8

9 Eyecup, USB stick

Step 9

10a Battery: XT30, pack, cap

Step 10 (body closed)

10b XT30 junction

Step 10

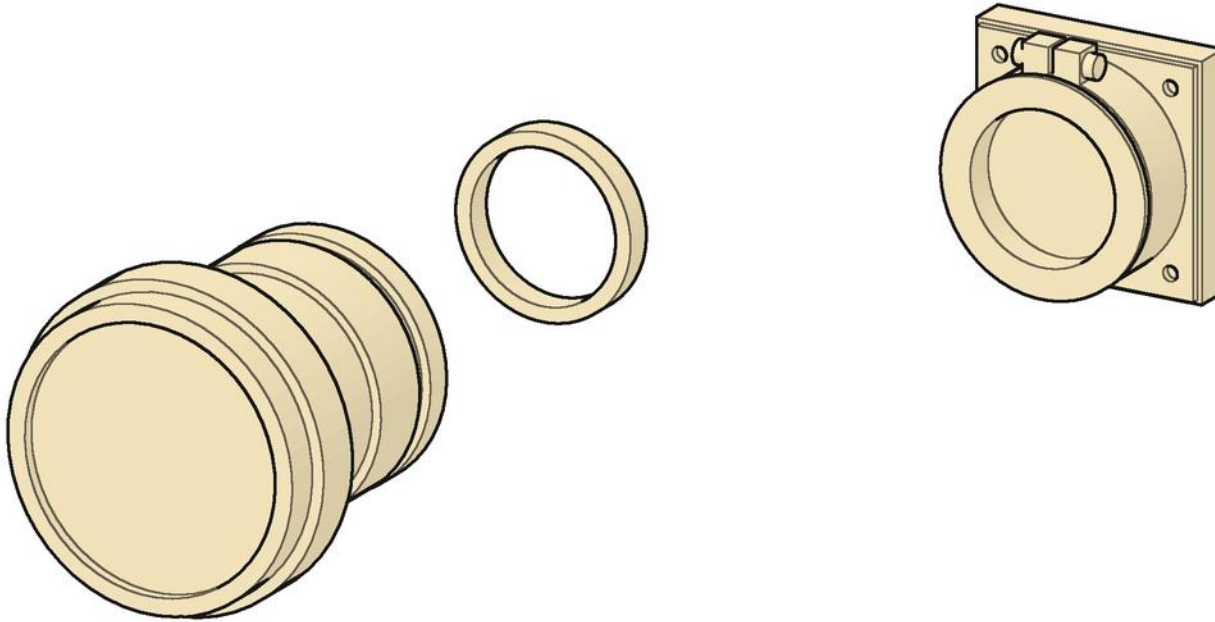
10c Power on, level check, gates

Step 10

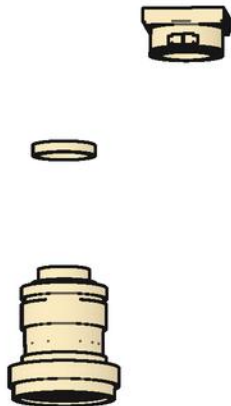
B0 Camera and lens on the bench

Bench, before any body part is printed (MP-CAM, G-CAM-1, G-LENS)

A from the front-left, above



B from above



C result, close-up, from the front-left, above



YOU NEED

- Raspberry Pi Global Shutter camera
- C-CS adapter ring 5 mm
- C-mount lens (Kowa LM6HC)
- Tools: (11) Paint pen, (12) ESD mat + wrist strap, (5) Multimeter

DO THIS

- 1 Measure and photograph the camera and lens as MEASURED-PARTS MP-CAM asks. Record the C-flange-to-cover-rear distance at $s = 0$ and at the final s .
- 2 Keep the tripod block on and clamp the camera to a bench tripod. Support the lens barrel in a padded V-block. Never let the lens hang on the camera alone.
- 3 Seat the C-CS adapter firmly in the back-focus ring (BFAR). Draw one paint-pen line across adapter and BFAR.
- 4 With the official Pi supply and an external monitor, set back focus with the lens at infinity (target more than 50 m away). Lock the BFAR split-tab screw. Record s ; it must lie between 0 and 3.0.
- 5 Remove the lens while the camera stays supported. Only then remove the tripod block and bag its 2 screws.

CAUTION

Plug the camera FPC only with the power off.

Never screw the BFAR out past $s = 3.0$. If infinity needs more, stop: record both stops and update CAM before printing.

If the tripod block will not come off cleanly, stop: the camera model must be updated first.

CHECK

- ☐ s recorded, inside 0-3.0
- ☐ paint mark across adapter and BFAR
- ☐ tripod block off, 2 screws bagged

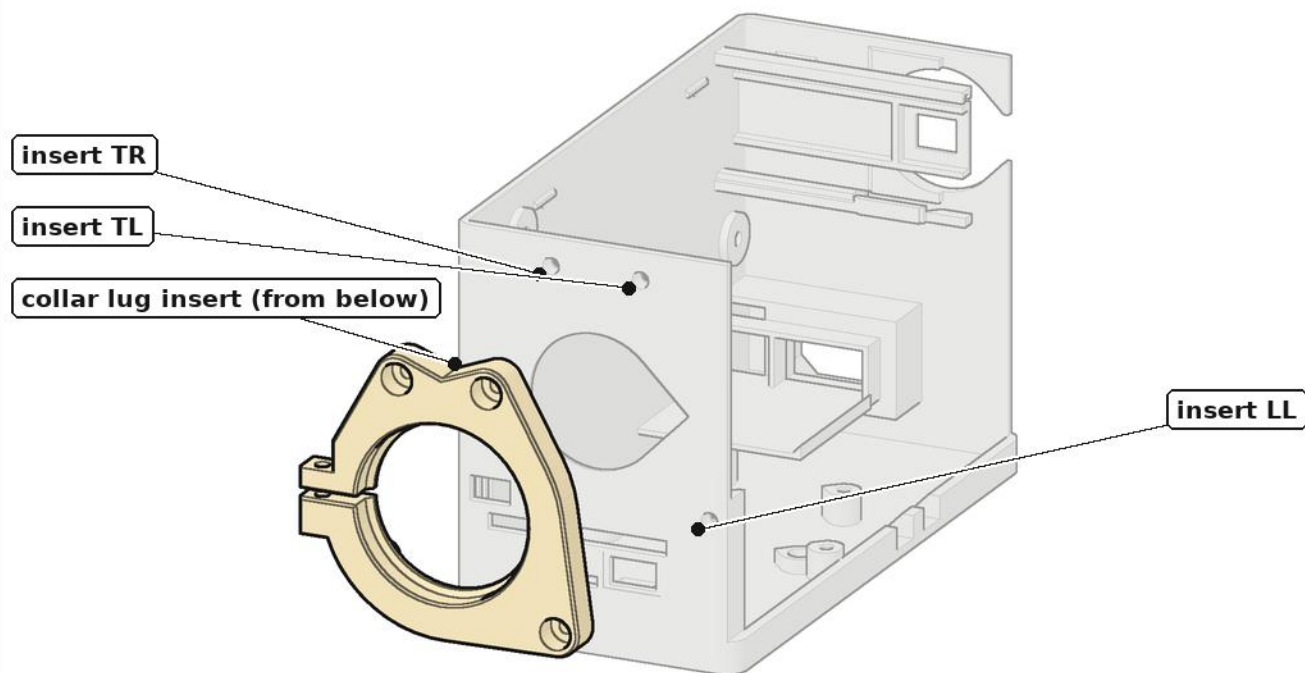
TIPS (physical blocker check)

BX-12 (buying tool 13): handle dia 30 or less; with a 1/4 in hex holder use a PH1 bit of 50 mm or more whose turned shank (6.5 or less) covers at least the first 8 mm.

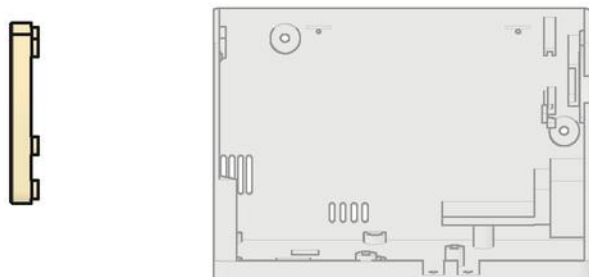
B1 Heat-set inserts

Bench, tub empty, any time before step 3 (collar insert: before step 7)

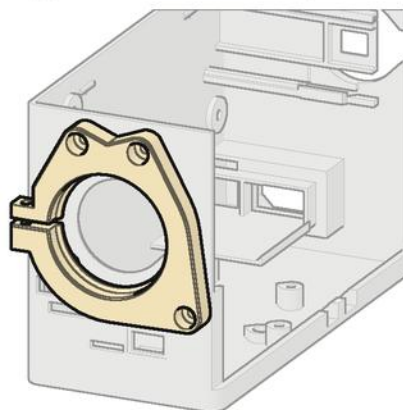
A from the front-left, above



B from the left



C result, close-up, from the front-left, above



YOU NEED

- tub (printed)
- lens collar (printed)
- Bag D: 3 x M3 brass heat-set insert, short
- Bag E: 1 x M3 brass heat-set insert, L 5.7
- Tools: (2) Temperature-controlled soldering iron + solder

DO THIS

- 1 Heat the iron with the M3 heat-set tip.
- 2 Press 3 short inserts (bag D) into the front-wall bosses TL, TR and LL from the front, square to the face, finishing 0.1 mm below it.
- 3 Back each boss with a block through the open left side: the bosses hang behind a 2.5 mm wall. Go slowly.
- 4 Press 1 long insert (bag E, L 5.7) into the lens collar lower lug from below.
- 5 Let all inserts cool. Check that an M3 screw turns in by hand.
- 6 Clear the 3 sacrificial 0.2 mm membranes in the collar with a 3.4 mm hand tool; remove the debris.

⚠ CAUTION

A crooked insert is the usual failure in a small ASA boss: check squareness on every one.

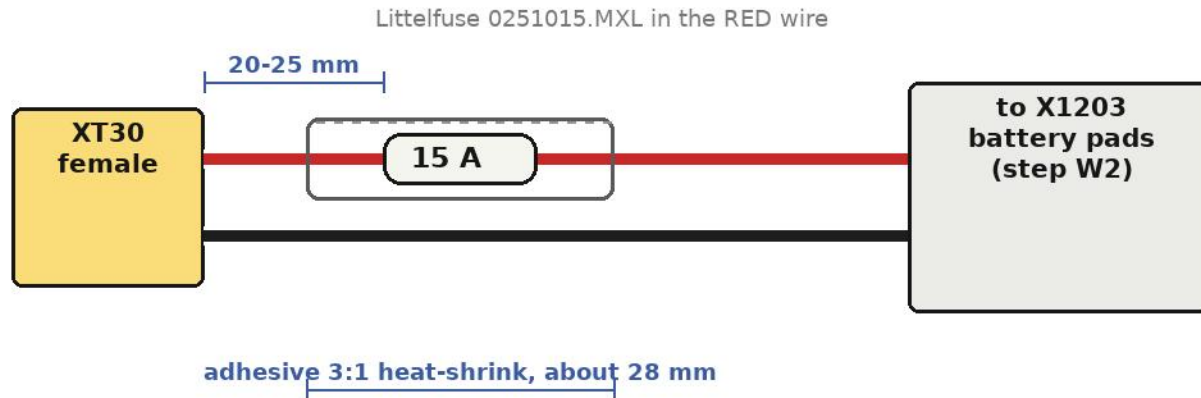
Do not change the collar washer seats, cone or foot faces while clearing the membranes.

CHECK

- ☐ 4 inserts square and flush
- ☐ M3 screw turns in by hand in each
- ☐ insert maker, OD, length recorded

1a Fuse the battery pigtail

Step 1, bench wiring (WIRING s7)



Fuse leads: trim each to 6.5 mm = 1.5 mm stub at the fuse body + 5 mm lap joint on the stripped red conductor.

Solder each lead by hand at 350 C for 5 s at most (datasheet). Never re-solder the XT30 with the fuse fitted.

Meter: XT30 (+) to the far end of the red wire: less than 0.1 ohm.

YOU NEED

- XT30 pigtail, 18 AWG silicone, cut to L_{cut} (XT30U female)
- fuse 15 A Littelfuse 0251015.MXL
- adhesive heat-shrink 3:1
- Tools: (2) Temperature-controlled soldering iron + solder, (3) Wire stripper + flush cutter, (4) Heat gun, (5) Multimeter

DO THIS

- 1 Cut the pigtail to L_{cut} = 200 mm + the G-W2 pads-to-W length, rounded up to 5 mm (200 mm if the pads are at W). L_{cut} runs from the pad joint to the XT30 rear and includes the fuse splice.
- 2 If the pads-to-W length is more than 25 mm, stop: do not cut. The battery-lead drop would pass 0.15 V (gate G-W5); route the pair under the board straight to the hole, move W, or use 16 AWG, and re-check.
- 3 Strip the red conductor about 20-25 mm from the XT30 female.
- 4 Trim each fuse lead to 6.5 mm. Lap 5 mm of lead on the stripped conductor; leave a 1.5 mm stub at the fuse body.
- 5 Solder each lead (350 C, 5 s at most).
- 6 Shrink one adhesive 3:1 sleeve (about 28 mm) over the whole splice.
- 7 Meter the red path end to end: less than 0.1 ohm.

CAUTION

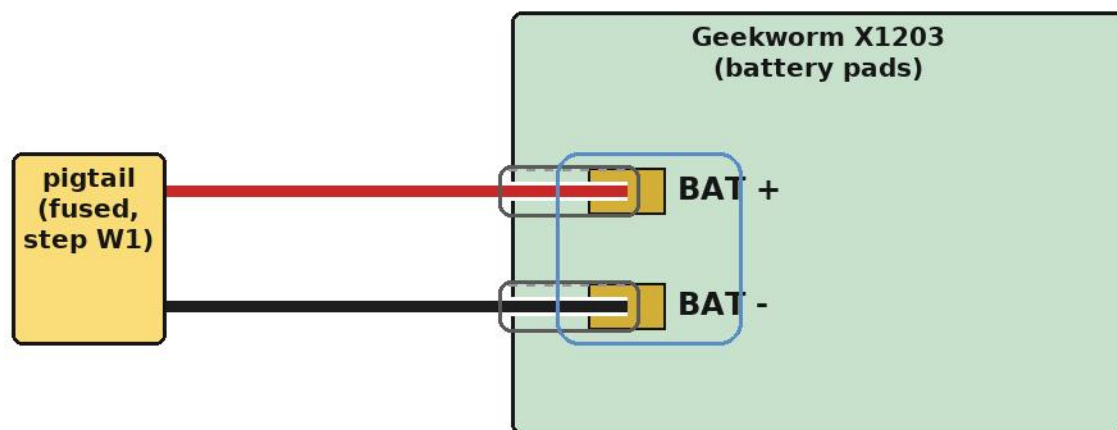
Never re-solder the XT30 female once the fuse is fitted.

CHECK

- ☐ red path < 0.1 ohm
- ☐ splice fully covered

1b Pigtail to the X1203 battery pads

Step 1, bench wiring



RTV bead over both joints (strain relief); no tie

Red to +, black to -. Adhesive heat-shrink over both joints, then the neutral-cure RTV bead. The U at the W mark and the taped leg are on the next steps of this page.

The pack is **NEVER** connected at the bench. Meter + to - on the pigtail: no short, before anything else.

YOU NEED

- Geekworm X1203
- fused pigtail (1a)
- Kapton tape, 3 strips
- neutral-cure silicone RTV (not acetox)
- Tools: (2) Temperature-controlled soldering iron + solder, (3) Wire stripper + flush cutter, (4) Heat gun, (5) Multimeter, (12) ESD mat + wrist strap

DO THIS

- 1 Solder red to the + battery pad and black to the - pad.
- 2 Shrink adhesive heat-shrink over both joints.
- 3 Mark W with a pen on the port edge, 31.5 mm from the button-end edge.
- 4 Lead the pair on the top face to W, straight in from inside the board ($y < 23$), not along the port edge: the keeper fingers kf3 and kf4 sit on that edge.
- 5 Put one Kapton strip over the board edge at W. Fold the pair round the edge. Tape the leg flat under the board with 2 Kapton strips, straight in from W for about 32 mm, to above the pigtail hole. Keep the pair within 1 mm of the mark. The rest hangs free.
- 6 Pads on the underside: run the pair straight under the board to that point and tape it, with no U.
- 7 Strain relief: a neutral-cure silicone RTV bead over both joints and the first 8-10 mm of the pair, bonded to the board. Keep it off the pogo pads and the connectors.
- 8 Meter + to - on the pigtail: no short.

CAUTION

The pack is **never** connected at the bench.

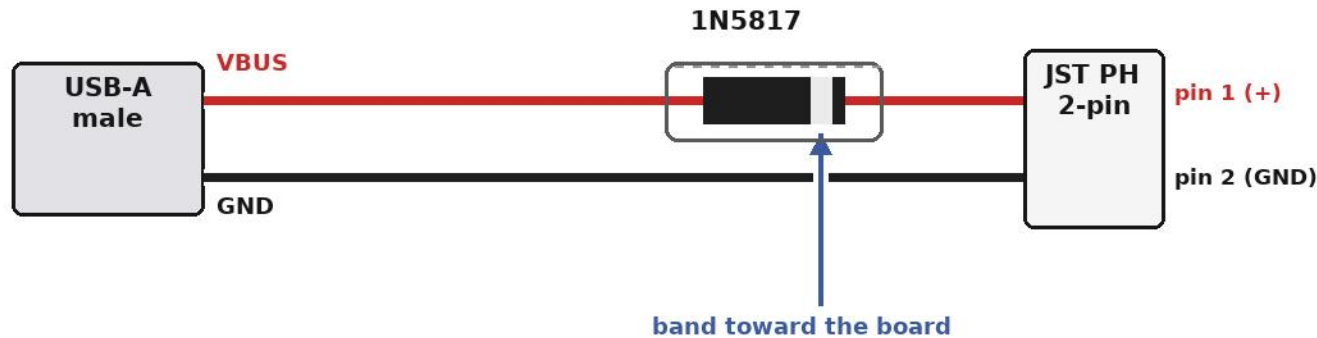
No tie to a standoff: the RTV bead is the strain relief. Never acetox RTV (it corrodes).

CHECK

- ☐ no short + to -
- ☐ joints insulated
- ☐ U at the W mark (within 1 mm), leg taped flat
- ☐ RTV over both joints and 8-10 mm of the pair

1c EVF 5 V lead with diode

Step 1, bench wiring



The diode goes in the + conductor, band (cathode) toward the PH plug, i.e. toward the EVF board. Adhesive heat-shrink over the diode and both joints.

Meter (diode test): USB-A VBUS (+ probe) to PH pin 1 reads a forward drop; reversed probes read open. USB-A GND to PH pin 2: continuity. No continuity between the two conductors.

YOU NEED

- USB-A to 2-wire lead, 200 mm class
- 1N5817 Schottky diode
- JST PH 2-pin plug
- adhesive heat-shrink
- Tools: (2) Temperature-controlled soldering iron + solder, (3) Wire stripper + flush cutter, (4) Heat gun, (5) Multimeter

DO THIS

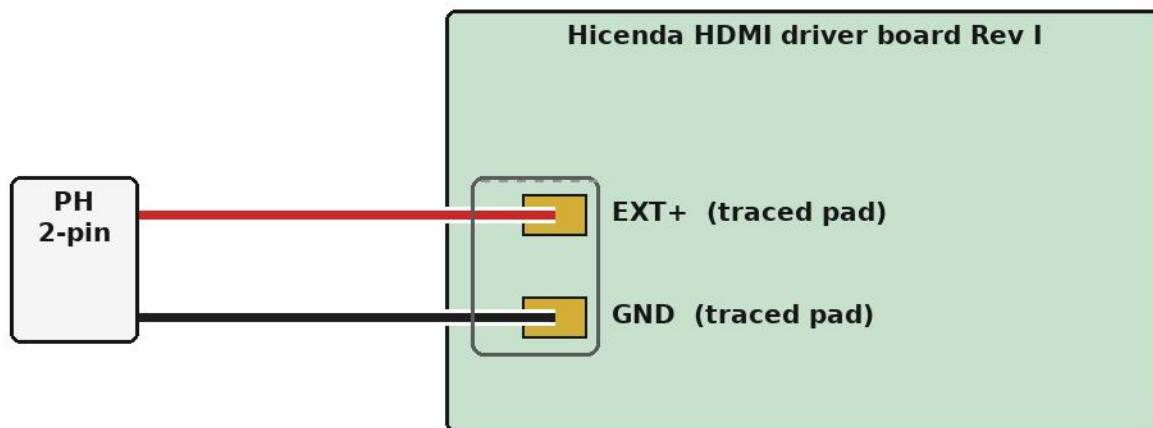
- 1 Cut the + conductor near the free end. Solder the 1N5817 into it, band (cathode) toward the board end.
- 2 Fit the JST PH 2-pin plug on the free end: + on pin 1, GND on pin 2.
- 3 Shrink adhesive heat-shrink over the diode and both joints.
- 4 Meter: VBUS to PH pin 1 forward through the diode; GND to pin 2.

CHECK

- ☐ diode band toward the PH plug
- ☐ polarity metered

1d EVF board pigtail

Step 1, bench wiring



First photograph both sides of the board (EVF gate G1). Then solder the PH 2-pin pigtail to the traced EXT+ and GND pads; heat-shrink as strain relief.

Keep the pin order of the EVF 5 V lead (step W3): + on PH pin 1, GND on pin 2, so the junction mates pin for pin.

YOU NEED

- Hicenda HDMI driver board Rev I
- JST PH 2-pin pigtail
- Tools: (2) Temperature-controlled soldering iron + solder, (4) Heat gun, (12) ESD mat + wrist strap

DO THIS

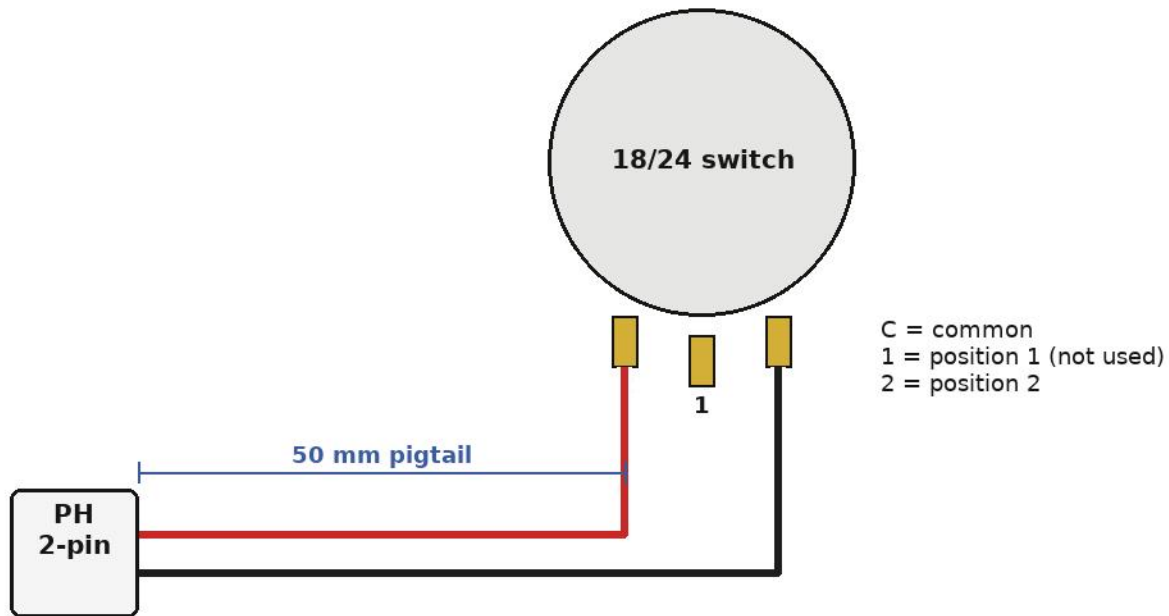
- 1 Photograph both sides of the board (EVF gate G1).
- 2 Solder the PH 2-pin pigtail to the traced EXT+ and GND pads.
- 3 Add heat-shrink strain relief.

CHECK

- ☐ photos taken
- ☐ pin 1 = EXT+

1e 18/24 switch pigtail

Step 1, bench wiring



Solder the 50 mm JST PH pigtail to the common lug and the position-2 lug (not lug 1). Heat-shrink both joints. It is a plain contact, so either conductor may go on either lug. Position 2 closed = GPIO13 low = 24 fps; open = 18 fps.

YOU NEED

- mini 2-position rotary switch
- JST PH 2-pin pigtail, 50 mm
- Tools: (2) Temperature-controlled soldering iron + solder, (3) Wire stripper + flush cutter, (4) Heat gun

DO THIS

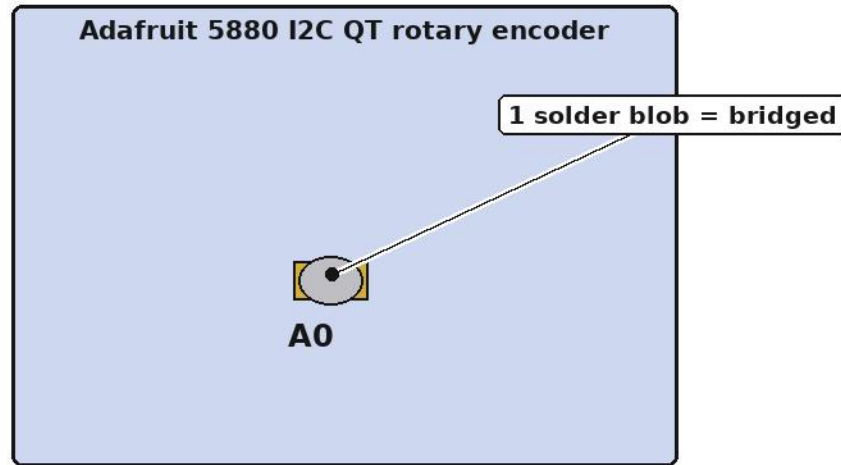
- ❶ Solder the 50 mm JST PH pigtail to the common lug and the position-2 lug.
- ❷ Heat-shrink both joints.

CHECK

- ☐ lug 1 left free
- ☐ both joints insulated

1f Encoder address, QT lead splice

Step 1, bench wiring



Bridge the A0 jumper with one solder blob. The encoder then answers at I2C address 0x37. The default 0x36 would clash with the X1203 fuel gauge.

YOU NEED

- Adafruit 5880 I2C QT rotary encoder
- QT lead parts: Adafruit 4397 (sockets) + Pololu 5521 (JST-SH), 4 thin sleeves, 1 adhesive sleeve
- Tools: (2) Temperature-controlled soldering iron + solder, (3) Wire stripper + flush cutter, (4) Heat gun, (5) Multimeter, (12) ESD mat + wrist strap

DO THIS

- 1 Bridge the A0 jumper with one solder blob (address 0x37).
- 2 QT lead (WIRING W-QT): map continuity on both leads first. 4397: SH pin 1 GND black, 2 3V3 red, 3 SDA blue, 4 SCL yellow. Pololu 5521: black is pin 1, find the other three with the meter.
- 3 Join by SH pin number, never by colour alone. Joints 91-115 mm from the socket tip, 8 mm apart, one thin sleeve each, one adhesive sleeve (about 35 mm) over all.
- 4 Finished length 250 +/- 10 mm, socket tip to SH plug face.

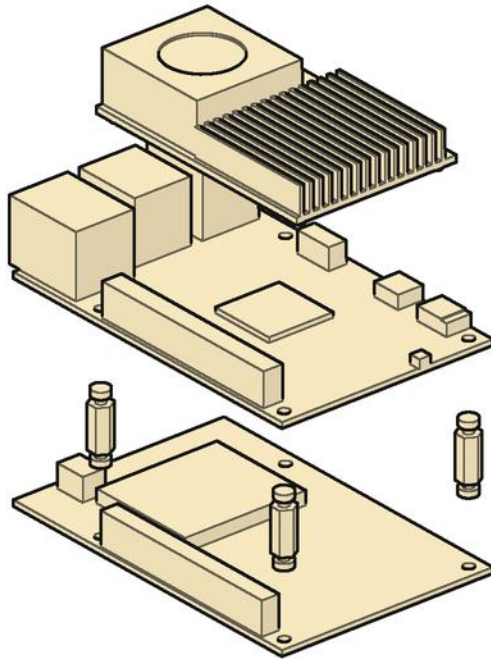
CHECK

- ☐ A0 bridged
- ☐ QT splice: 4 joints, pin 1 to pin 1, continuity end to end, no short
- ☐ no joint or sleeve outside 85-121 mm from the socket tip
- ☐ QT lead 240-260 mm

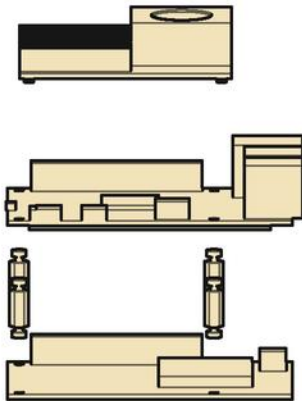
1g Bench stack: X1203 + Pi 5 + cooler

Step 1, bench sub-assembly (joins the body at step 4)

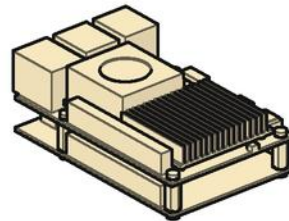
A from the front-right, above



B from the left



C result, close-up, from the front-right, above



YOU NEED

- X1203 (with pigtail)
- X1203 kit: 4 standoffs, 8 M2.5 screws
- Raspberry Pi 5
- Active Cooler
- microSD card 32 GB
- Bag F: 1 x X1203 kit: 4 standoffs + 8 M2.5 screws
- Tools: (1) PH1 screwdriver, (12) ESD mat + wrist strap
- Torque: kit M2.5 screws 0.2 N m

DO THIS

- 1 Stack only when the RTV bead (1b) is tack-free. Do not pull on the pigtail before the full cure the tube states.
- 2 Stack the X1203 and the Pi 5 with the kit: 4 standoffs, 8 M2.5 screws, 0.2 N m, metal to PCB only.
- 3 Fit the Active Cooler (pins seated). Plug its fan lead into the Pi FAN header; route it as the cooler ships.
- 4 Flash the microSD (gate G-W3). Then take the card OUT of the Pi: it must stay out until step 5.

CAUTION

The microSD hits the front wall on the stack path if it is left in.

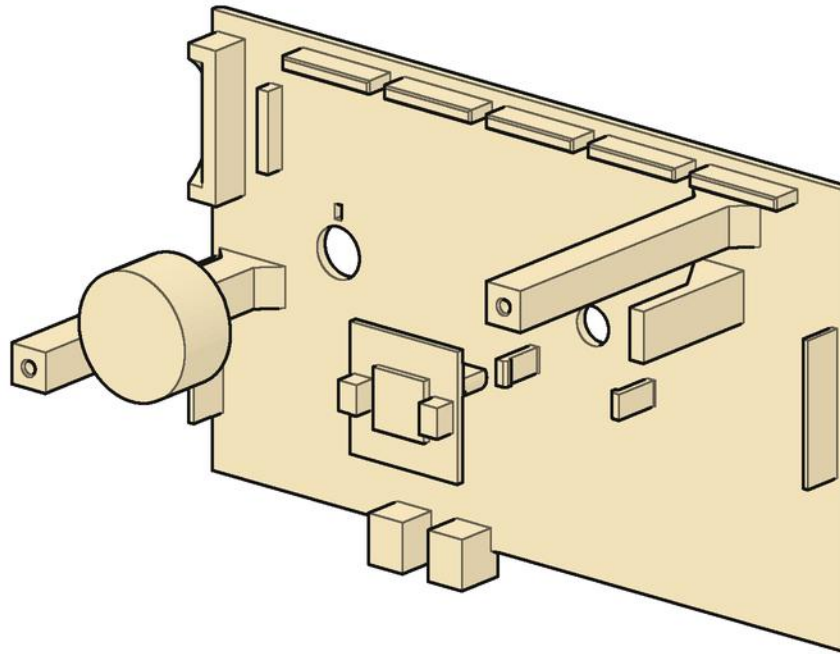
CHECK

- ☐ stack square, cooler pins seated
- ☐ fan lead plugged
- ☐ pigtail leg still taped, RTV tack-free
- ☐ microSD out

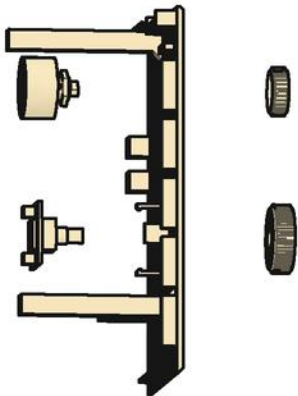
2 Bench: panel

Step 2, bench sub-assembly (joins the body at step 8)

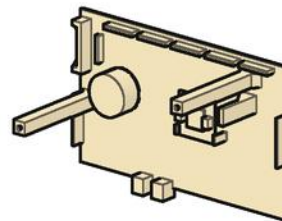
A from the front-right, above



B from above



C result, close-up, from the front-right, above



YOU NEED

- panel (printed, satin silver)
- encoder (A0 bridged)
- 18/24 switch + nut + washer
- 2 knobs (printed)
- Tools: (11) Paint pen, (6) 12.7 mm, (7) Junior hacksaw + small flat file + caliper with depth rod + soft-jaw vise, (16) Printer paper
- Torque: switch nut finger-tight + 1/8 turn (about 0.3 N m)

DO THIS

- 1 Fill the engraving with the paint pen.
- 2 Switch shaft, off the panel: fit the 18/24 switch dry (tab in its slot, washer as supplied, nut finger-tight). Measure how far the shaft stands above the panel face (P, caliper depth rod). Take the switch off.
- 3 Clamp the shaft in soft vise jaws right next to the cut, the switch hanging free. Cut P - 6.9 mm off the shaft end and deburr. Never clamp the bushing thread; never saw with the switch on the panel.
- 4 Refit the switch, tab in its slot, nut finger-tight + 1/8 turn.
- 5 Snap the encoder into its cradle by pushing on the back of its board until all 3 hooks click.
- 6 Lay the 2 paper strips on the panel face round the encoder shaft. Thumb flat on the back of the encoder board, press knob_exp onto its D shaft with the other palm until it stops on the paper. Press knob_fps on the same way, D flat to the shaft flat. Pull the paper out.

⚠ CAUTION

From now on the knobs ride with the panel: never press a knob while the panel is on the body.

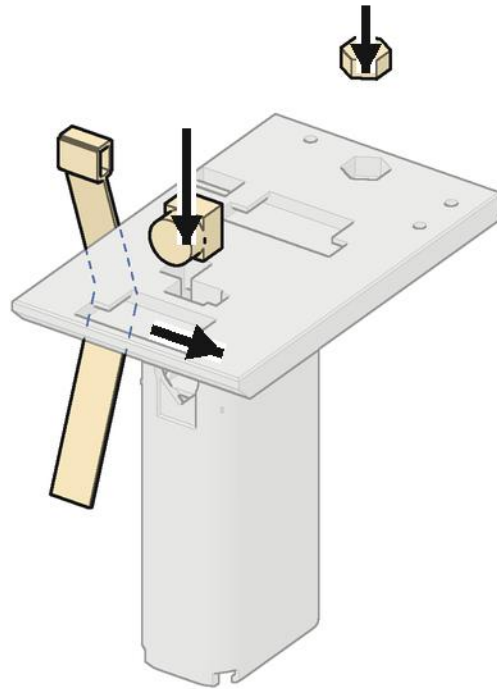
CHECK

- ☐ encoder under both hooks, shaft square
- ☐ switch tab in its slot
- ☐ switch shaft 6.7-7.1 above the face
- ☐ knobs 0.2 off the panel (paper drags)
- ☐ knob flat lines up with the 18 and 24 ticks at the stops

3a Base: tripod nut, run button, strap

Step 3

A from the front-left, above



YOU NEED

- base + grip (printed, black)
- run button (pre-wired, red cap)
- 12 mm hand strap
- Bag G: 1 x 1/4-20 UNC hex nut, steel
- Tools: (10) Flat bar or steel rule, (8) Fine tweezers

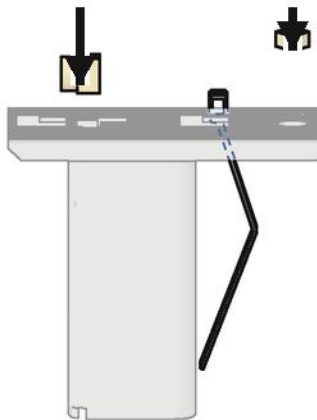
DO THIS

- 1 Press the tripod nut into its pocket from the top with a flat bar until it sits flush.
- 2 Before the run button goes in: pass the run-lead socket end in through the base opening and push it up through the base passage from below (nothing is under the passage yet).
- 3 Drop the run button into its cradle through the base opening, drawing the slack up through the passage. Press its red cap on through the grip-face hole.
- 4 Thread the strap.

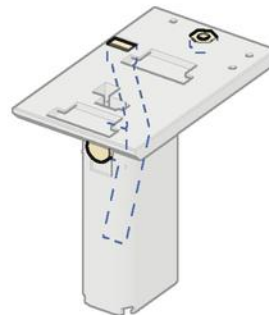
CHECK

- ☐ nut flush in its pocket
- ☐ run cap 2.5 mm proud of the pad

B from the left



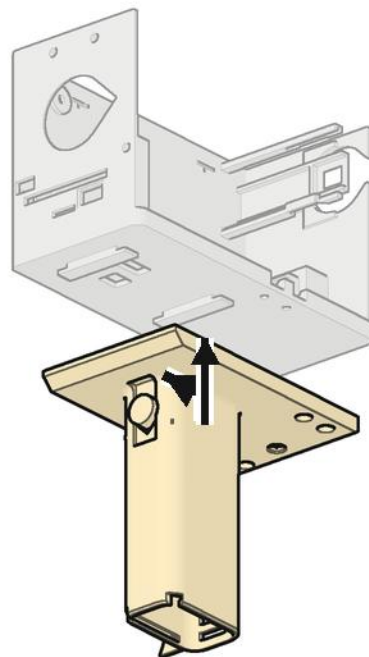
C result, close-up, from the front-left, above



3b Tub onto the base, lock screw s_j

Step 3

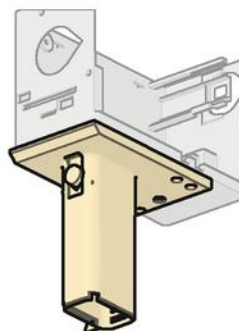
A from the front-left, below



B from the front



C result, close-up, from the front-left, below



YOU NEED

- tub (printed, satin silver)
- base assembly (3a)
- Bag A: 1 x PT 3.0 x 12 thread-forming screw for plastics, pan head, PH1
- Tools: (1) PH1 screwdriver, (13) Adjustable torque screwdriver 0.1-0.6 N m, (8) Fine tweezers
- Torque: s_j 0.35-0.5 N m, stop at head contact

DO THIS

- 1 Hold the tub over the base with the base 10 mm to the rear. Push the run-lead sockets (separate 1-pin housings, already up through the base passage) on up through the rear (-X) end of the floor run-lead hole into the tub.
- 2 Lower the tub, tongues through the windows. Keep the lead lightly pulled up so it stays in the rear end of the hole.
- 3 Slide the base 10 mm forward until the tongues sit under the lips. The lead moves to the front end of the hole by itself.
- 4 Drive s_j up through the base counterbore into the tub floor boss (straight PH1, from below).
- 5 Lay the run lead in its floor lane and tape it flat with Kapton. Let the socket end hang out over the open +Y side.

⚠ CAUTION

Thread the run lead before the tub goes down. Do not fish it up after the slide.

CHECK

- ☐ base slid the full 10 mm, does not lift off
- ☐ s_j at head contact (tally 1)
- ☐ run lead up through the -X end of the floor hole, not over the base top
- ☐ run lead not pinched between base and floor; taped flat in its lane

TIPS (physical blocker check)

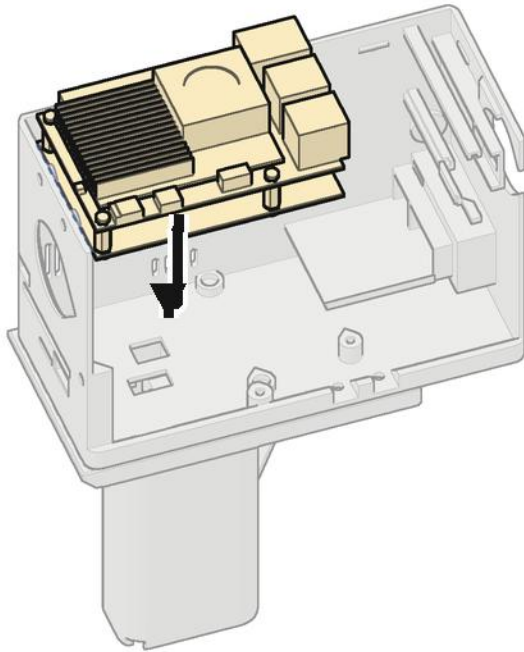
Thread the run-lead sockets up through the floor hole before you lower the tub; keep the lead in the rear (-X) end of the hole while the tub drops and slides.

BX-12: hold the strap aside for the s_j driver handle (2.35 mm clearance).

4a Lead routing, then the Pi stack

Step 4 (ESD strap; microSD out)

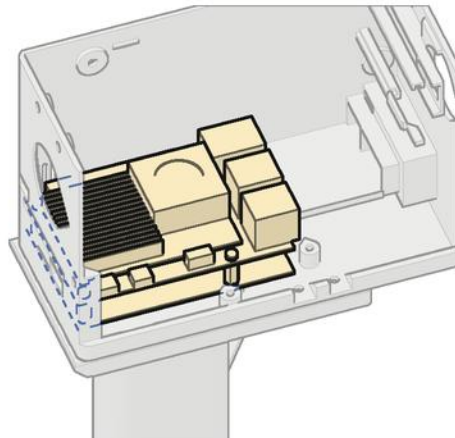
A from the left, above



B from the front



C result, close-up, from the left, above



YOU NEED

- bench stack (1g)
- Tools: (12) ESD mat + wrist strap

DO THIS

- 1 Check that the run lead is still taped flat in its lane, its socket end held out over the open +Y side.
- 2 Feed the XT30 end of the pigtail down through the pigtail hole into the grip (long side front to back); the leg under the board comes down with the stack. Once the stack is down, push the XT30 and the free pigtail up into the empty grip bay: nothing may hang below the grip mouth until step 10.
- 3 Hold the stack's port edge away from the U: more than 10 mm from the W mark.
- 4 Lower the stack about 3 mm behind the front wall and 2 mm off the right wall.
- 5 Below the right-wall pads, move it against the right wall.
- 6 Just above the floor, slide it 2.8 mm forward and set it down on its 4 bosses. Nothing clicks: the kit screw heads drop into the boss pockets.

⚠ CAUTION

Nothing may lie under the stack: run lead taped in its lane, pigtail leg taped under the X1203, XT30 down its hole.

CHECK

- ☐ stack flat on its 4 bosses, no rock
- ☐ pigtail leg still taped flat under the X1203
- ☐ U at the W mark, not under the keeper finger kf4 or the bar (x -41 and beyond)

TIPS (physical blocker check)

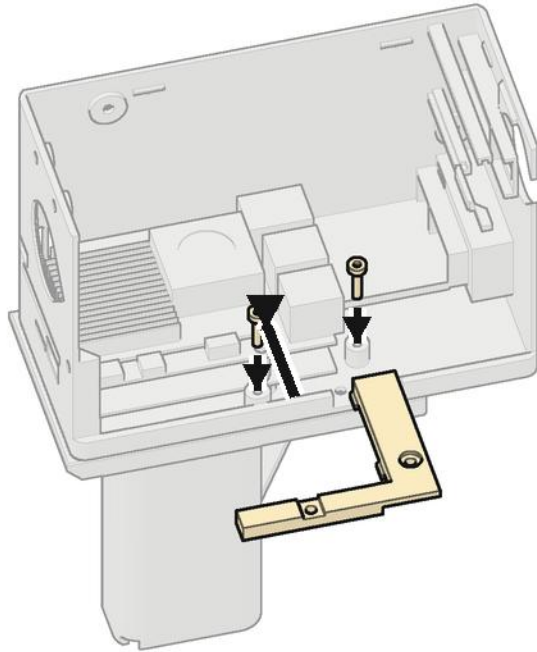
C-2: feed the XT30 with its long side front-to-back; once it is in the bay, steer it rearward (below z -20 the run-button cradle wall is in the way).

The pigtail leg is already taped under the X1203 (page 1b): just feed the XT30 down the hole as the stack comes down. Check that the run lead is still taped flat in its lane.

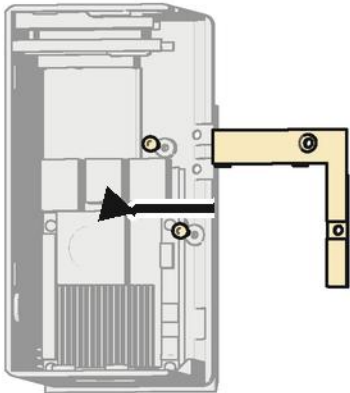
4b Pi keeper, screws s_k1 and s_k2

Step 4

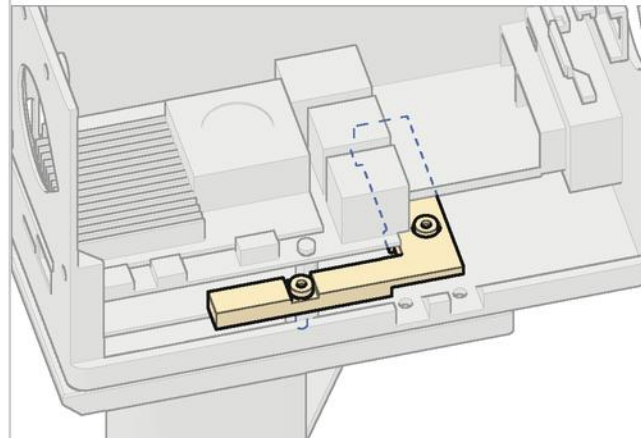
A from the left, above



B from above



C result, close-up, from the left, above



YOU NEED

- Pi keeper (printed, black)
- Bag A: 2 x PT 3.0 x 12 thread-forming screw for plastics, pan head, PH1
- Tools: (1) PH1 screwdriver, (13) Adjustable torque screwdriver 0.1-0.6 N m
- Torque: s_k1, s_k2 0.35-0.5 N m, stop at head contact

DO THIS

- 1 Hold the keeper level from the open left side, 0.5 mm above its 2 bosses and 1.2 mm behind its place.
- 2 Slide it in (toward the right wall) under the stick-guide rail until its port fingers are over the X1203 edge.
- 3 Push it 1.2 mm forward: the 2 USB fingers go under the Pi.
- 4 Lower it onto its bosses. Drive s_k1 and s_k2 straight down.

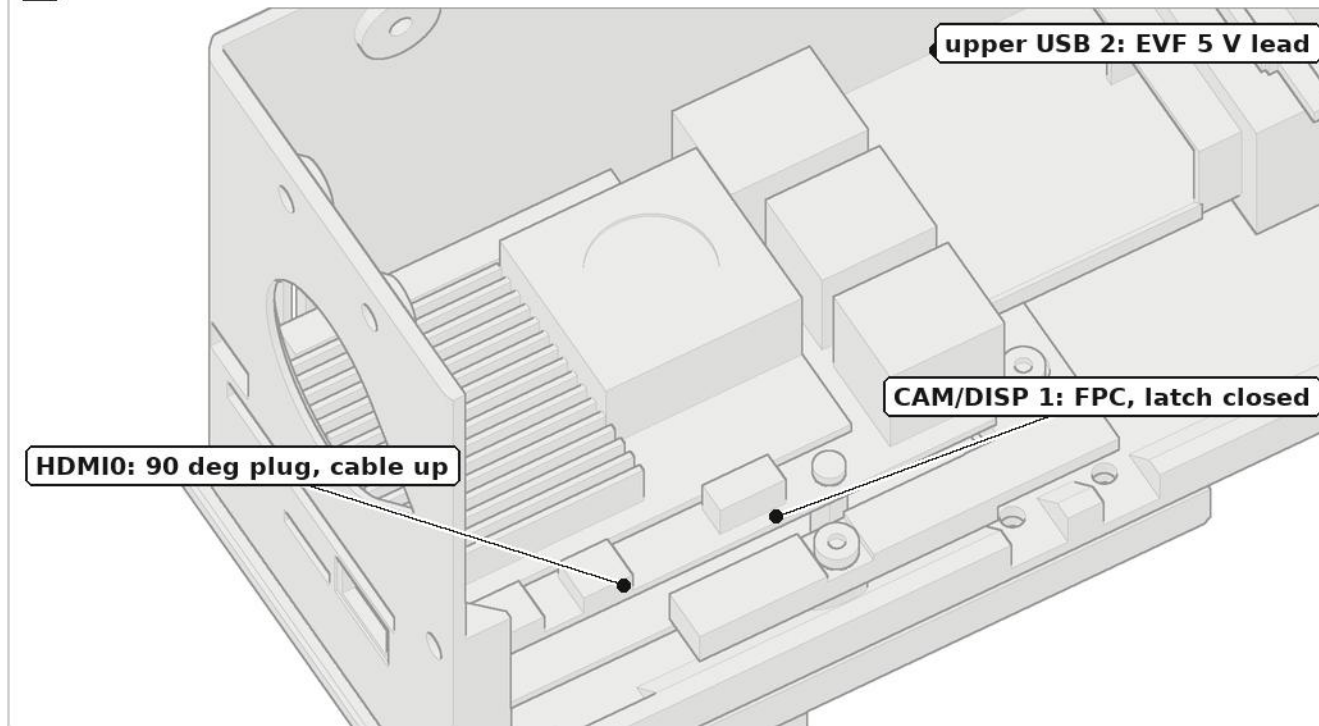
CHECK

- ☐ keeper fingers over the X1203 at all 4 stations
- ☐ keeper flat on its 2 bosses
- ☐ s_k1, s_k2 at head contact (tally 1 each)

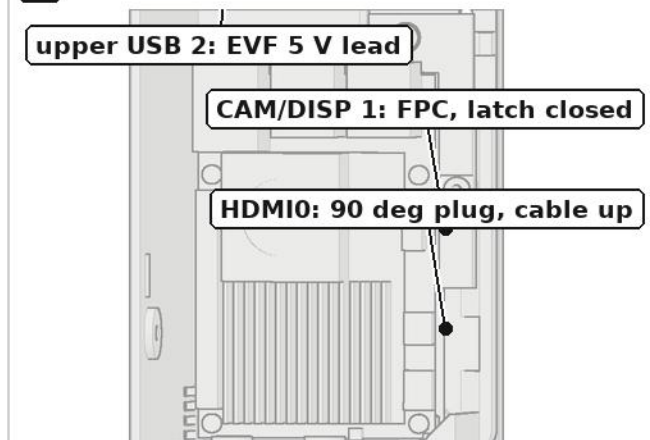
4c Plug the Pi end of HDMI, FPC and EVF 5 V lead

Step 4 (top open)

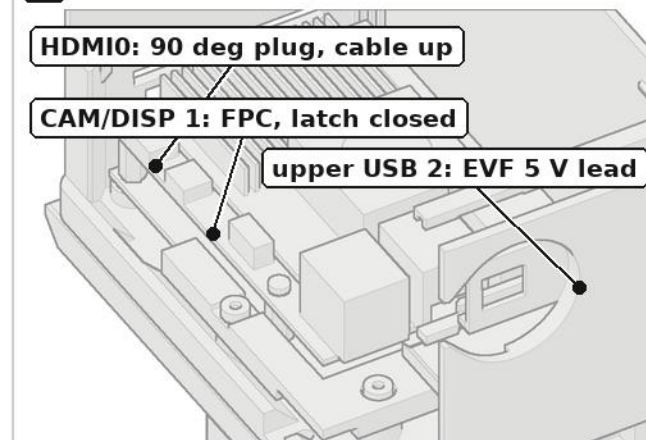
A from the front-left, above



B from above



C from the rear-left, above



YOU NEED

- micro-HDMI 200 mm (90 deg Pi plug)
- camera FPC 22-to-15, 200 mm
- EVF 5 V lead (1c)
- Tools: (12) ESD mat + wrist strap

DO THIS

- 1 micro-HDMI: 90 deg plug into HDMI0 (the one nearest the USB-C), cable leaving upward. Leave the EVF end loose.
- 2 FPC: into CAM/DISP 1, contacts as printed on the cable; close the latch. Leave the camera end loose.
- 3 EVF 5 V lead: USB-A into the upper USB 2 port. Leave the PH end loose.

CHECK

- ☐ 3 plugs fully home at the Pi

4d Header leads

Step 4 (top open, header in sight)

Pi 5 header J8 seen from above; board edge (even pins) at the top



PIN 1: inner row, at the end farthest from the USB ports

● **QT lead:** red 1 (3V3), blue 3 (SDA), yellow 5 (SCL), black 6 (GND)

● **18/24 lead:** GPIO13 on 33, GND on 34

● **run lead:** GPIO26 on 37, GND on 39

⚠ **Count from pin 1. Pins 2 and 4 carry 5 V: one pin off and 5 V reaches the encoder 3V3 wire. Photograph the header before the body is closed.**

YOU NEED

- QT lead, 250 mm (Adafruit 4397 sockets spliced to Pololu 5521 JST-SH, splice W-QT made at the bench)
- 18/24 lead (Dupont to PH, 200 mm)
- run lead (from step 3)
- Tools: (8) Fine tweezers, (12) ESD mat + wrist strap

DO THIS

- 1 Plug one housing at a time, top open, header in sight.
- 2 QT lead on pins 1/3/5/6: red 1, blue 3, yellow 5, black 6.
- 3 18/24 lead on 33/34: GPIO13 on 33, GND on 34.
- 4 Run lead on 37/39: GPIO26 on 37, GND on 39.
- 5 Seat each housing straight down with open tweezer tips straddling the wire, pressing on both sides of the housing top, until it stops on the header plastic. Fingers do not fit beside them. Then tug each wire straight up gently: no housing may lift.
- 6 Run the QT and 18/24 leads along the right wall and across behind the blower inlet. Keep the QT splice sleeve in the straight part of the cross run, not in a corner. Park their free ends out of the open left side.
- 7 Lay the run lead over the cooler shroud, below z 40.

⚠ CAUTION

Count from pin 1. A housing one row over puts 5 V from pin 2 on the QT 3V3 wire.

Single 1-pin housings only (a 1x2 only on 33/34 or 37/39). Never a 1x3 or 2x3 shell.

CHECK

- ☐ QT red on pin 1, 18/24 on 33/34, run on 37/39
- ☐ 8 contacts, all housing tops level, each wire tugged
- ☐ header photographed
- ☐ QT splice sleeve in the straight cross run
- ☐ leads along the right wall, run lead over the shroud

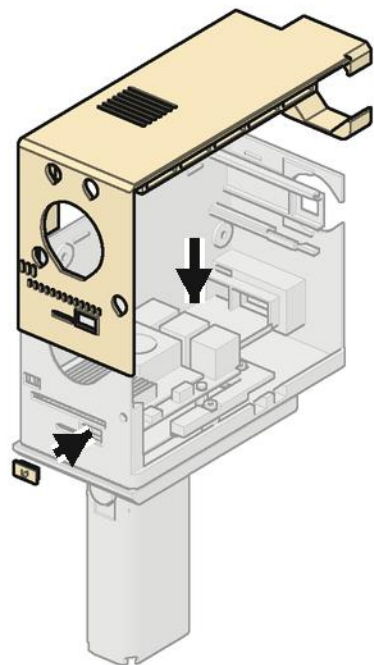
TIPS (physical blocker check)

Seat each socket straight down with open tweezer tips straddling the wire (both sides of the top), then tug each wire. Single sockets only (a 2-way only on 33/34 or 37/39).

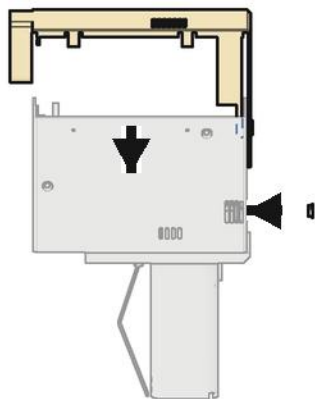
5a Plunger and hood

Step 5

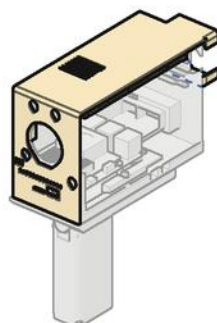
A from the front-left, above



B from the right



C result, close-up, from the front-left, above



YOU NEED

- plunger (printed, black)
- hood (printed, black)

DO THIS

- 1 Hold the plunger in the front-wall hole, flange outside.
- 2 Lower the hood straight down until all 4 hooks click. The front plate traps the plunger.

⚠ CAUTION

Never rock the hood in: straight down only.

CHECK

- ☐ hood level on the wall tops all round
- ☐ 4 hooks clicked
- ☐ plunger moves in and returns

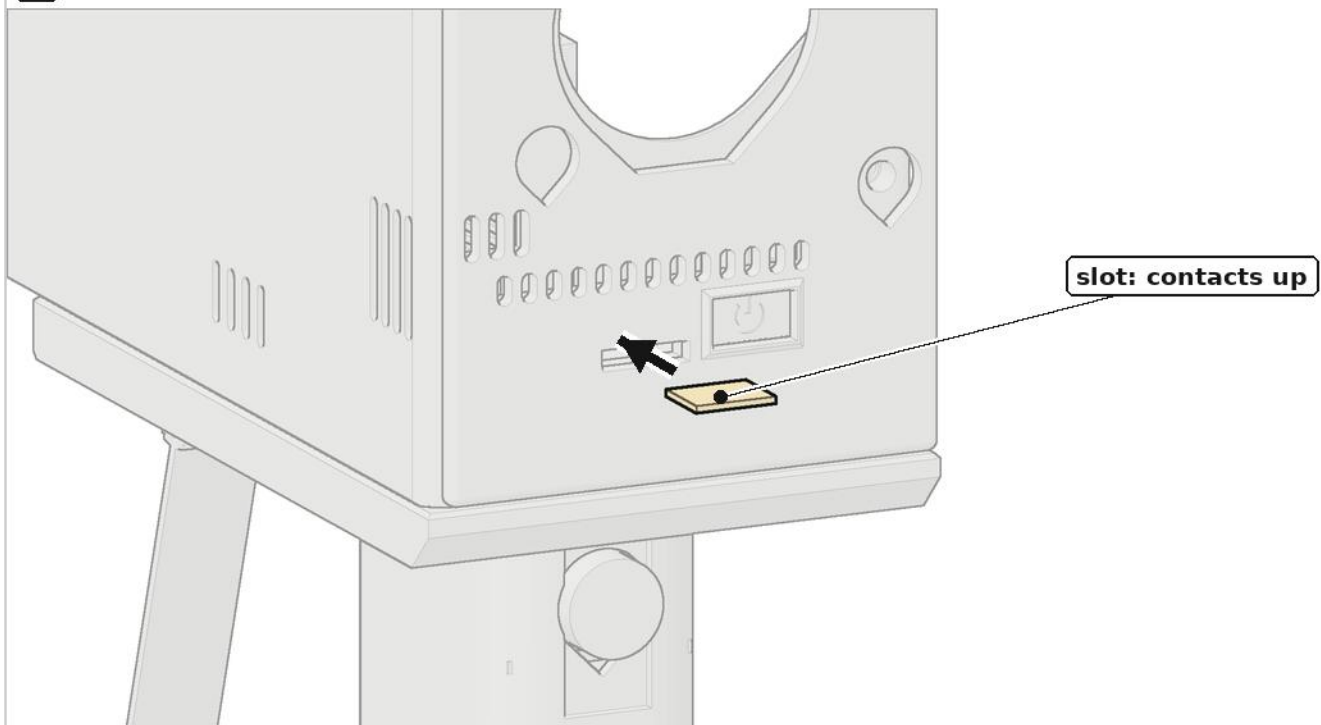
TIPS (physical blocker check)

BX-6: rest the body nose-up on its rear face so the plunger stays in its hole by gravity, then lower the hood straight down onto the body.

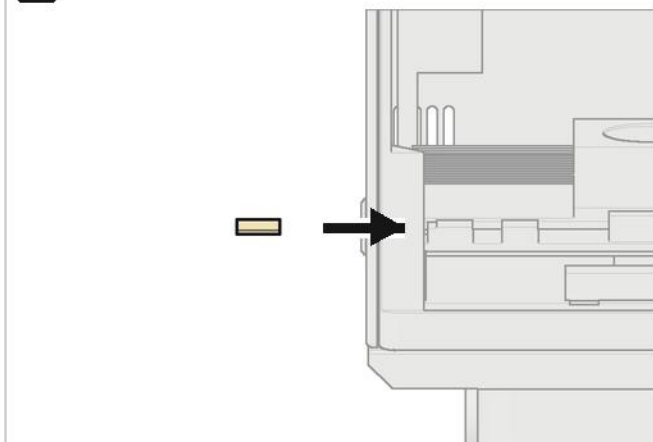
5b microSD through the front slot

Step 5

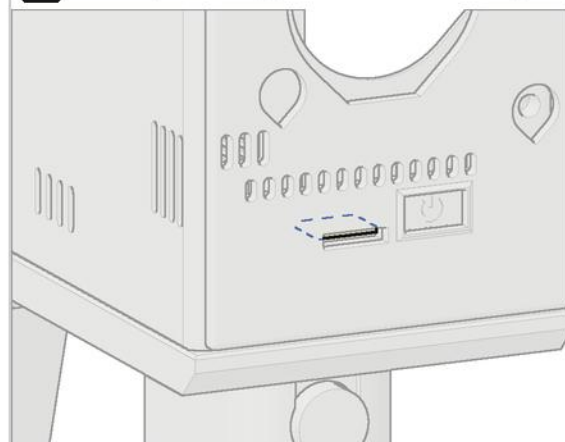
A from the front-right



B from the left



C result, close-up, from the front-right



YOU NEED

- microSD (flashed at 1g)
- Tools: (8) Fine tweezers

DO THIS

- 1 Push the microSD through the slot low in the front plate (both walls) with tweezers, contacts up, until it latches.

CHECK

- ☐ card latched, tip about 2.5 mm proud of the Pi edge, inside the slot

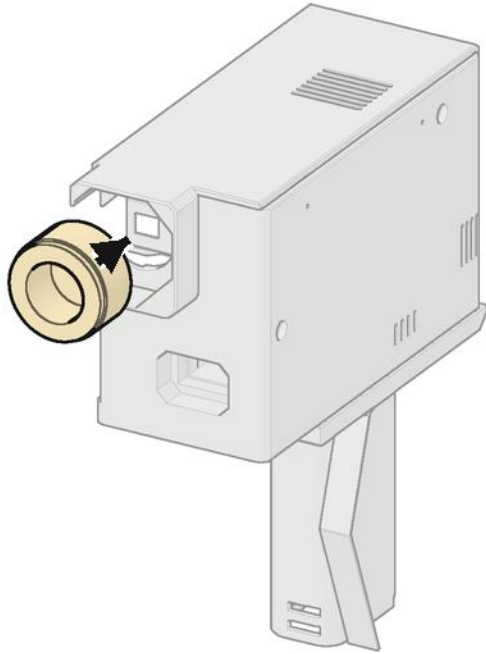
TIPS (physical blocker check)

C-14: check on receipt whether the Pi 5 slot is push-push. If it is, removal is push-to-release, not pull.

6a Eyepiece

Step 6

A from the rear-right, above



YOU NEED

- eyepiece 0PE039-16X

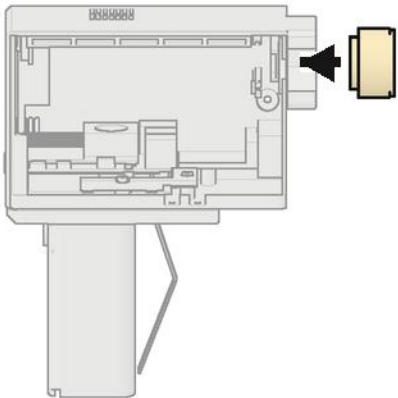
DO THIS

- 1 Push the eyepiece spigot forward into the rear-wall bore through the housing, flange on the rear face.

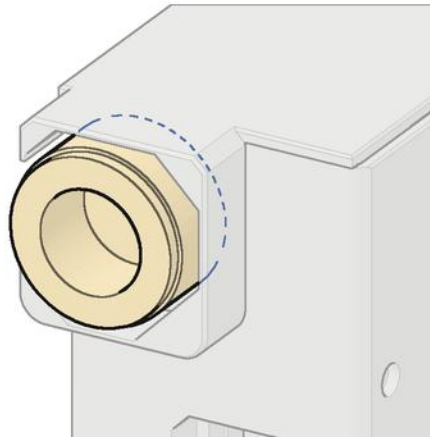
CHECK

- ☐ spigot flange flat on the rear face

B from the left

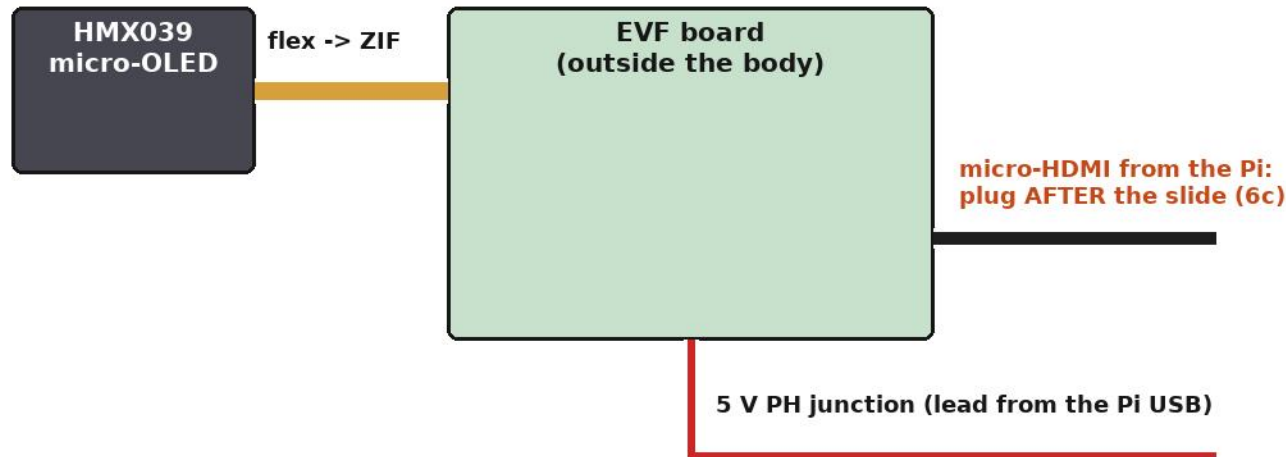


C result, close-up, from the rear-right, above



6b Mate the EVF leads outside the body

Step 6 (ESD mat; pack out)



- 1 Flex into the ZIF, contacts as the kit shows; close the latch.
- 2 Mate the EVF 5 V lead PH plug with the board pigtail (pin 1 to pin 1).
- 3 micro-HDMI: NOT here. It goes in after the slide, up through the rail gap (page 6c, BX-1).

Flex and 5 V PH are mated outside the body, on the ESD mat, with the pack out. The HDMI is plugged in place after the slide (page 6c).

YOU NEED

- HMX039 micro-OLED + 50 mm flex
- EVF board (1d)
- foam pad 16 x 14 x 1.0, adhesive one side
- Tools: (8) Fine tweezers, (12) ESD mat + wrist strap

DO THIS

- ① Stick the foam pad to the OLED back (adhesive side to the OLED, centred).
- ② Flex into the board ZIF, latch closed.
- ③ Pull the EVF 5 V lead end out of the open left side. Mate the 5 V PH junction with the pair held about 60 mm (a hand's width) out of the open left side.
- ④ Do NOT plug the HDMI yet: it goes into the board after the slide (page 6c).

⚠ CAUTION

Never mate any lead with the pack connected.

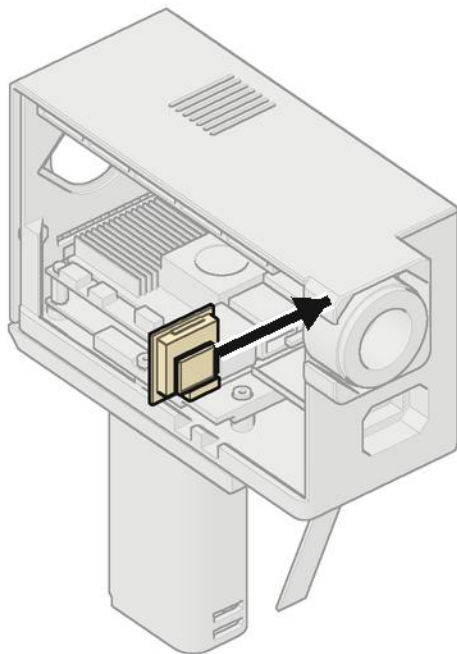
CHECK

- ☐ pad stuck flat on the OLED back
- ☐ flex square in the ZIF, latch closed
- ☐ PH fully home
- ☐ HDMI still loose

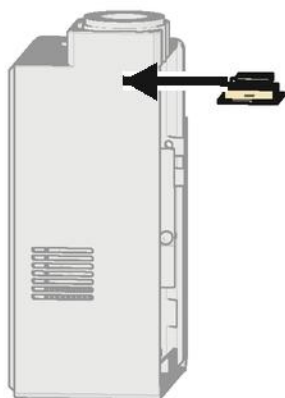
6c Slide the OLED and board in together

Step 6

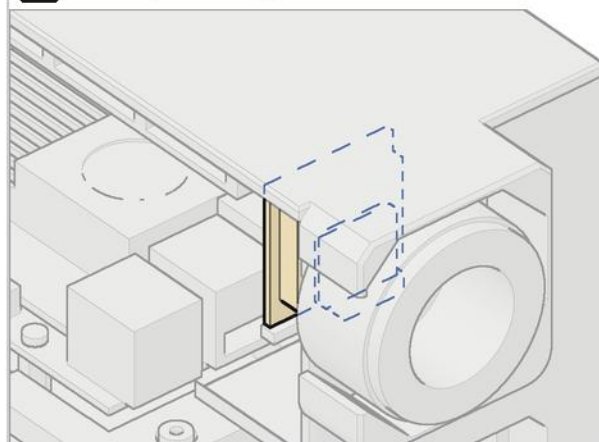
A from the rear-left, above



B from above



C result, close-up, from the rear-left, above



YOU NEED

- OLED + pad + board tethered pair (6b)
- Tools: (8) Fine tweezers, (10) Flat bar or steel rule, (12) ESD mat + wrist strap, (17) Inspection mirror

DO THIS

- 1 Slide the OLED and the board in together as a tethered pair until the board stops: OLED and pad into the cell, board into its slot.
- 2 Keep the PH junction riding above the bottom rail, beside the board. When the board is home, push the junction -Y and down into its space (ko_5v_end) with the 120 mm tweezers.
- 3 Only after gate G-W7 passed: pinch the right-angle HDMI plug's +Y end top and bottom (thumb under it), or hold it in the smooth-jaw pliers, cable leading -Y. Feed it in under the rail end, its bottom above the stick-guide top, lift it to the socket, then push it straight up: a fingertip flat on the stick-guide top, or straight up from the open well below (fingertip, or the steel rule held upright as a push stick).
- 4 Coil the HDMI slack over the stick guide (ko_hdmi_coil), clear of the cooler inlet.

⚠ CAUTION

Push the HDMI straight up. If it does not go, pull it back and re-aim; never lever against the board or the guide edge.

From now until the panel is on (step 8), keep the body level or nose-down: only the panel cap stops the eyepiece moving rearward.

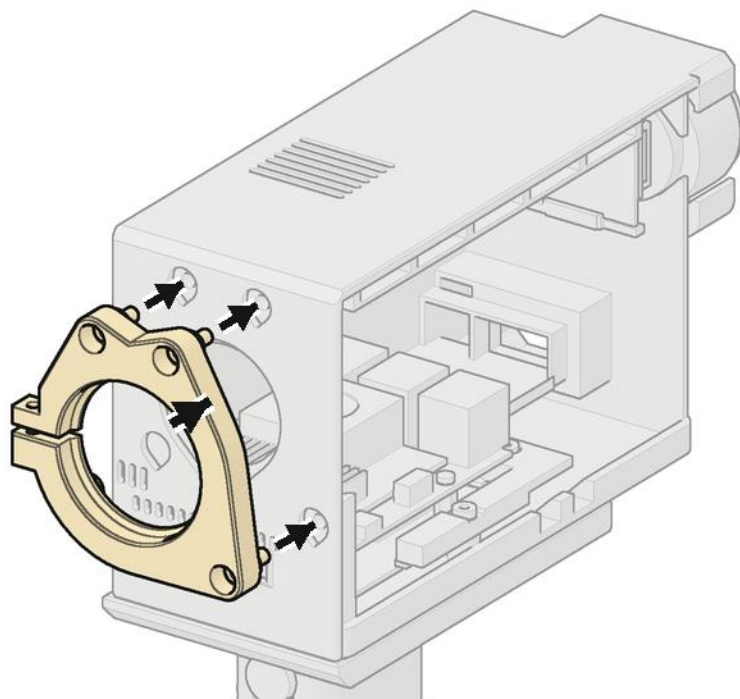
CHECK

- ☐ OLED in its cell, pad on its back
- ☐ board fully home
- ☐ junction in its space, clear of the rail
- ☐ HDMI fully home (mirror or fingertip)
- ☐ HDMI coil clear of the cooler inlet
- ☐ first print: gate G-EVF-3 (5 HDMI mate/unmate cycles) recorded

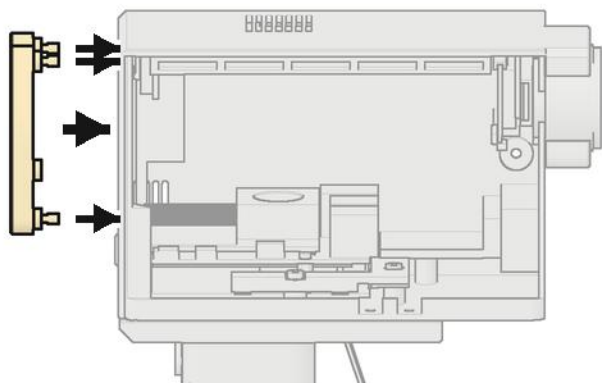
7a Lens collar, centred

Step 7 (B0 done; camera still out)

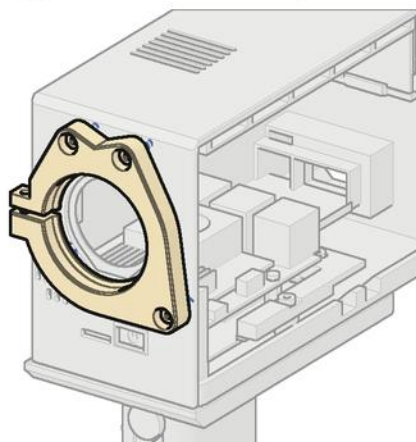
A from the front-left, above



B from the left



C result, close-up, from the front-left, above



YOU NEED

- lens collar (printed, black, with insert)
- centring gauge for this lens (tool 14)
- Bag B: 3 x M3 x 10 PH1 machine screw + flat washer
- Tools: (13) Adjustable torque screwdriver 0.1-0.6 N m, (14) Collar centring gauge
- Torque: s_c1, s_c2, s_c3 0.15 N m (provisional cap), stop at head contact

DO THIS

- 1 Put the collar feet through the 4 hood holes onto the tub face.
- 2 Start s_c1, s_c2, s_c3 with washers, 2 turns each.
- 3 Push the centring gauge through the collar until its rear cone seats in the tub lip and its front cone in the collar bore chamfer. Hold it home with a thumb.
- 4 Tighten s_c1, s_c2, s_c3 from the front on the torque screwdriver, 0.15 N m. Pull the gauge out.

⚠ CAUTION

Do this before the lens is fitted: with the lens on, the driver handle hits the lens barrel.

Never by hand: a hand-tight M3 is 2-3 times the value and can pull a short insert.

CHECK

- ☐ 4 feet flat through the hood holes
- ☐ gauge seated on both cones while tightening
- ☐ washers under heads

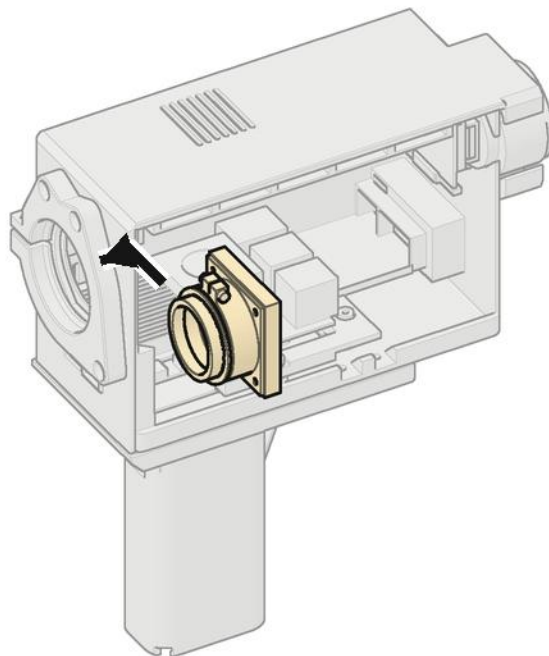
TIPS (physical blocker check)

BX-15: thumb on the upper right of the gauge face while driving s_c3; on its lower half while driving s_c1 and s_c2.

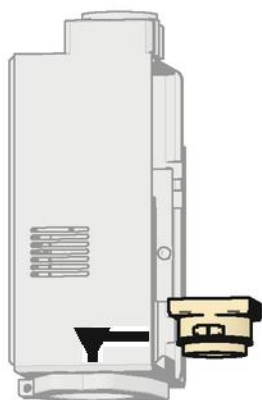
7b Camera and adapter into the cage

Step 7 (ESD strap)

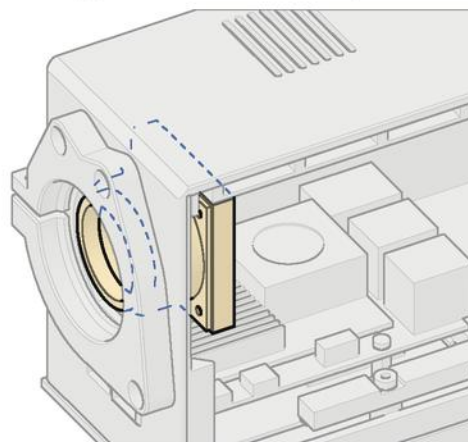
A from the front-left, above



B from above



C result, close-up, from the front-left, above



YOU NEED

- GS camera with adapter (B0)
- Tools: (12) ESD mat + wrist strap

DO THIS

- 1 Plug the FPC into the camera (15-pin end, latch closed).
- 2 Bring the camera in from the open left side, 2 mm high and 11 mm behind its place.
- 3 Lower it 2 mm, then push it 11.1 mm forward until the adapter has passed the tub lip and the lock tab has gone in between the two tab-catch tines under the hood. If the tab stops on a tine end, roll the camera level and push again.
- 4 It rests in its cage (BFAR in the counterbore, tab near the wall) until the lens carries it.
- 5 Fold the FPC slack into its loop space.

CHECK

- ☐ camera resting, not wedged (lifts and turns slightly)
- ☐ adapter mark lines up
- ☐ FPC folded, clear of the blower inlet

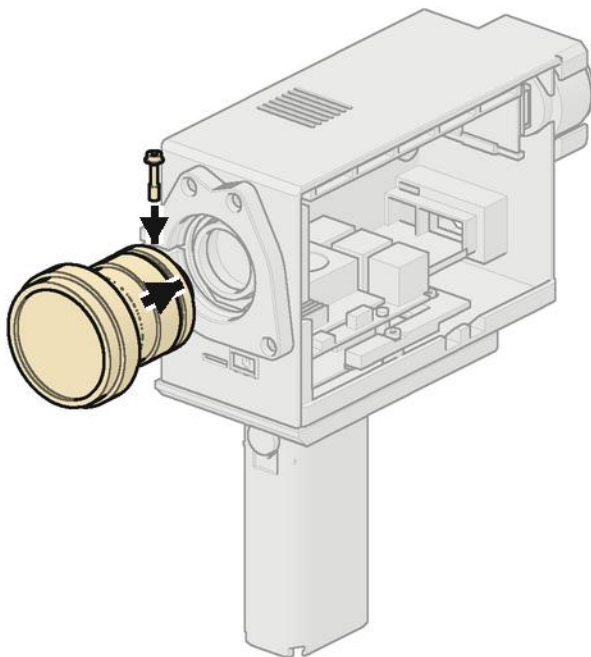
TIPS (physical blocker check)

BX-8: pre-fold the FPC S-fold on the bench (bend radius 1 mm or more, held with a Kapton strip) before the camera goes in; it must not touch the blower inlet.

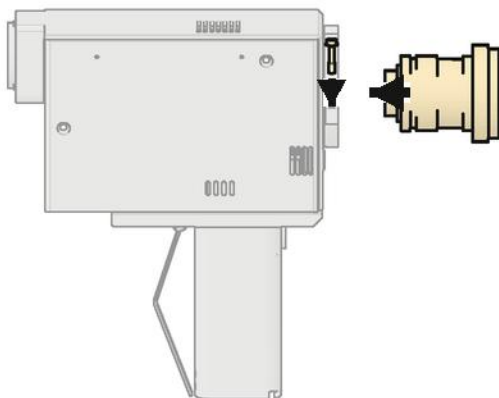
8a Lens through the collar, s_c4

Step 8 (panel still off)

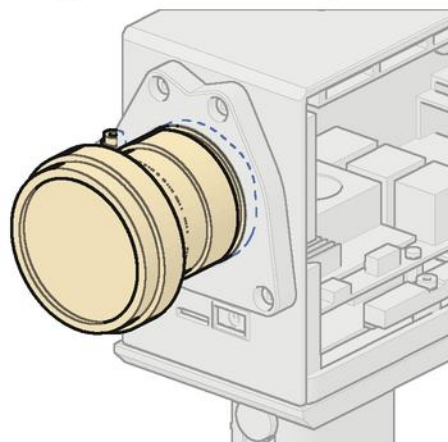
A from the front-left, above



B from the right



C result, close-up, from the front-left, above



YOU NEED

- C-mount lens (Kowa LM6HC)
- Bag C: 1 x M3 x 16 PH1 machine screw + flat washer
- Tools: (13) Adjustable torque screwdriver 0.1-0.6 N m
- Torque: s_c4 0.2 N m, stop at head contact

DO THIS

- 1 Fit s_c4 loosely. Lay the body on its right side on the folded cloth, open left side up.
- 2 From above, through the open left side, put one fingertip on the centre of the camera cover and press the camera forward (+X) onto its lip catch. Push only: never pinch or turn the cover.
- 3 Pass the lens through the collar. Screw it into the adapter with fingertips until it stops, with no extra snug. The camera turns with it about 3 deg until its metal lock tab meets the hood tab catch, which holds it through metal.
- 4 Set iris and focus; tighten the 2 thumb screws.
- 5 Push the lens gently rearward until the knurl seats on the collar cone.
- 6 Stand the camera upright, one hand on the grip until s_c4 is snug (it tips at about 6 deg if let go). With s_c4 still loose, turn lens and camera together the other way until the lock tab stops on the other tine (about 6.5 deg of free roll), then back about half way: the tab now floats between the tines and the camera hangs on the lens.
- 7 Snug s_c4 straight down from above: 0.2 N m on the torque screwdriver.

⚠ CAUTION

Never hold, pinch or turn the camera by the cover or the PCB: the tab catch takes the thread torque.

s_c4 is always the last operation on the lens.

CHECK

- ☐ knurl seated on the cone, no axial play
- ☐ camera touches neither the lip, the counterbore nor the tab-catch tines
- ☐ by feel before s_c4: a soft stop each way at about 3 deg, tab left mid-way
- ☐ s_c4 at 0.2 N m

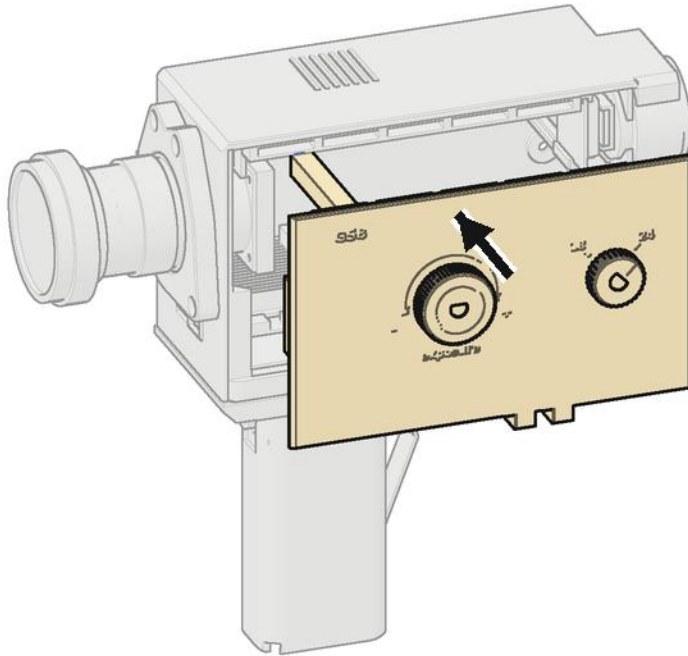
TIPS (physical blocker check)

Fingertip pushes the cover centre forward; turn the lens with fingertips until it stops. The camera stops on the tab catch. Never grip the cover. Before s_c4, turn lens and camera together the other way to the other tine, then back half way.

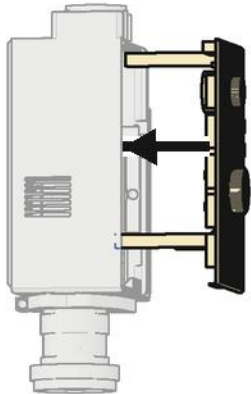
8b Panel leads, then the panel

Step 8

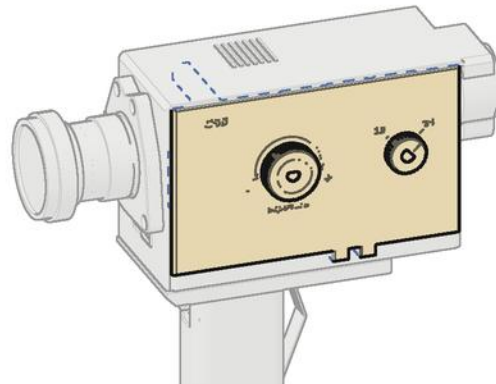
A from the left, above



B from above



C result, close-up, from the left, above



YOU NEED

- panel assembly with both knobs (step 2)

DO THIS

- 1 A helper holds the body upright by the grip until the panel is home (never leave it standing on its grip end: it tips at about 6 deg).
- 2 Prop the panel about a hand's width (60 mm) off the body, inner face toward it, its bottom edge on a block or book stack as tall as the grip (about 110 mm).
- 3 Mate the 18/24 PH junction first, with both hands, one on each housing. (The header ends went on at step 4.)
- 4 Then the QT: one hand steadies the panel, thumb behind the encoder's +X edge; the other holds the QT plug by its rear end and pushes it into the encoder's rear-side socket (the one nearest the lead), finishing with a fingernail on its back face. It is home when it stops (friction lock, no click).
- 5 Move the panel toward the body, body upright. At about 30 mm off, reach down from above between the body and the panel and lay the spare QT lead as a flat fold between the encoder and the 18/24 switch, on top of the HDMI cable. Lay the 18/24 spare and its PH junction in the same space at the switch end. Keep both away from the blower.
- 6 Keep the body upright until the panel is fully home.
- 7 Push the panel on toward the right: tongue into the hood groove, boss tabs into the lip notches, flush. The keeper finger becomes the camera rear catch.

CHECK

- ☐ QT plug fully seated at the encoder before the panel moves in
- ☐ before the last 30 mm: QT fold, 18/24 spare and PH junction between encoder and switch on the HDMI run; nothing over the blower inlet
- ☐ panel closes without pressure, flush with the hood band and the tub
- ☐ no lead pinched at the panel edge

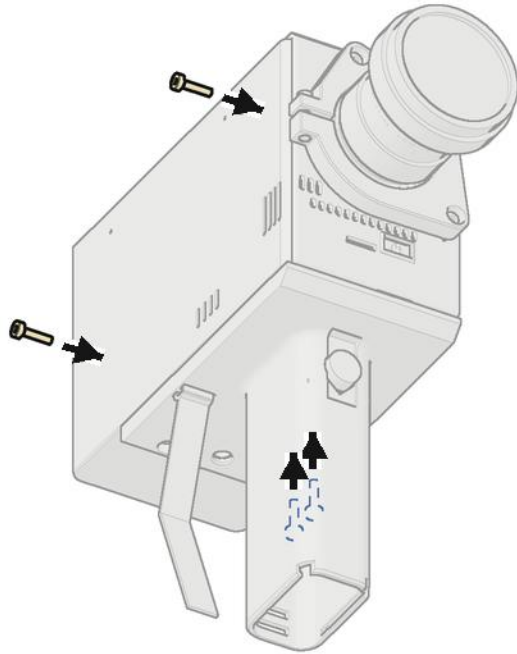
TIPS (physical blocker check)

A helper holds the body. Prop the panel a hand's width off on a block. Mate the small PH junction first, then push the QT plug in with a fingernail. Fold both spare leads and the small junction between the dial and the switch before the panel closes.

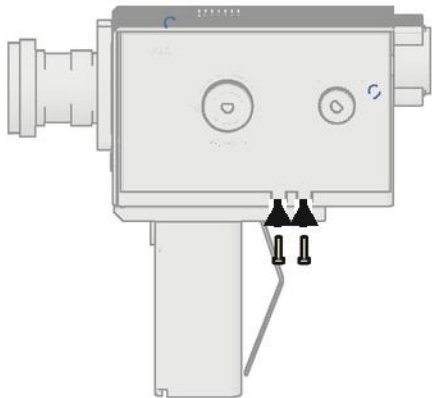
8c Four panel screws

Step 8

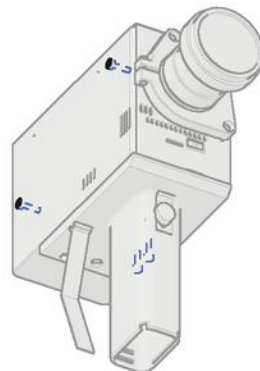
A from the front-right, below



B from the left



C result, close-up, from the front-right, below



YOU NEED

- Bag A: 4 x PT 3.0 x 12 thread-forming screw for plastics, pan head, PH1
- Tools: (1) PH1 screwdriver, (13) Adjustable torque screwdriver 0.1-0.6 N m
- Torque: 0.35-0.5 N m, stop at head contact

DO THIS

- 1 Hold the panel home and turn the camera over onto its hood roof on a folded cloth at the bench edge, grip up, right side toward you, its right face within about 20 mm of the edge (the driver handle and your hand go past the edge).
- 2 Lay a flat block no taller than 25 mm (a closed paperback) on the cloth against the left side, between the eyepiece and the lens, stood against a wall or a heavy object so it cannot slide. It bears on the hood band and the panel strip between the knobs and the cloth and keeps the panel home.
- 3 Drive s_b1 and s_b2 first (now downward) through the 2 base counterbores into the panel bosses.
- 4 Then drive s_r1 and s_r2 level from the right through the wall pads into the panel posts, the block taking the push; hold the grip with your other hand.
- 5 Turn the camera upright, a hand on the grip; set it down only on its right side.

CAUTION

Never lay the camera on its left side (the exposure knob would carry it), and never stand it on its grip end (it tips at about 6 deg).

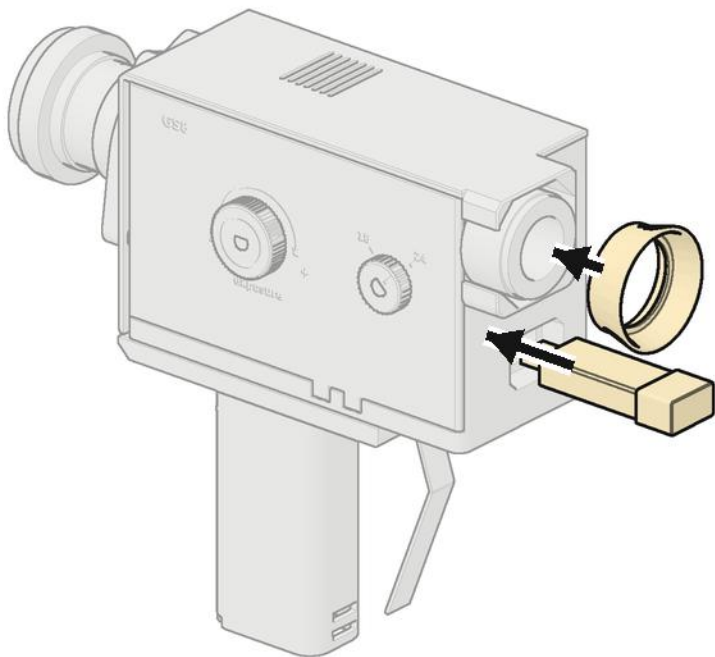
CHECK

- ☐ 4 screws at head contact, none stripped (tally 1 each)
- ☐ the dials turn; nothing rubs inside

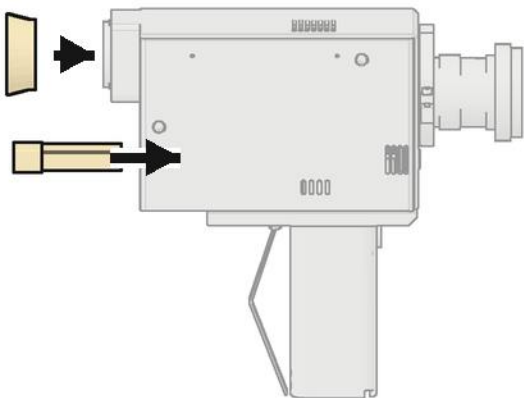
9 Eyecup, USB stick

Step 9

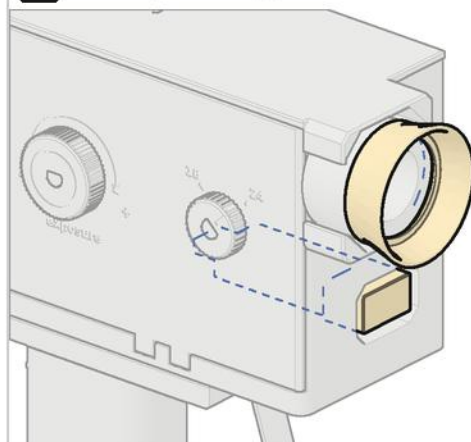
A from the rear-left, above



B from the right



C result, close-up, from the rear-left, above



YOU NEED

- eyecup (TPU)
- USB SSD stick + sleeve (printed)

DO THIS

- 1 Push the eyecup over the eyepiece barrel.
- 2 Fit the sleeve to the stick. Push the stick in from the rear into the lower USB 3 port until the sleeve is flush with the rear face.

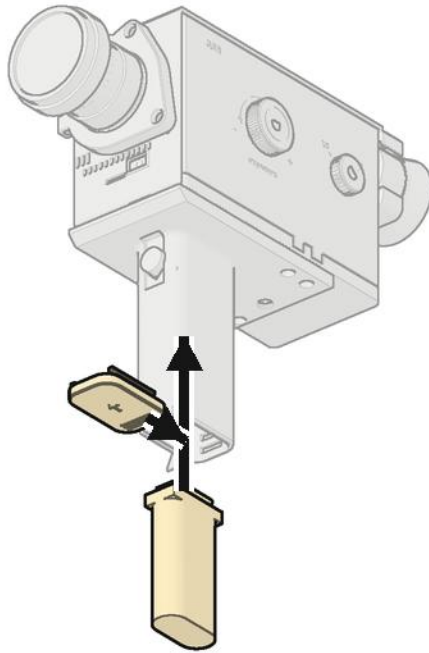
CHECK

- ☐ eyecup over the eye-end body
- ☐ sleeve flush

10a Battery: XT30, pack, cap

Step 10 (body closed)

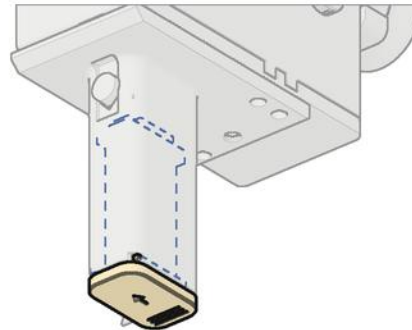
A from the front-left, below



B from the front



C result, close-up, from the front-left, below



YOU NEED

- 1S2P 18650 pack (charged out of the camera)
- cap (printed)

DO THIS

- 1 Lay the camera on its right side on the folded cloth (never on its left side; not on its hood roof now: the eyecup stands proud of it).
- 2 Draw the pigtail XT30 out of the grip mouth until the whole XT30 housing and at least 15 mm of wire behind it are out (estimate: about 28 mm of wire or more). If it does not reach, stop: never mate it inside the bay. Re-check the cut length (G-W2).
- 3 Hold the pack in your palm, its XT30 in your fingers, and plug it into the pigtail XT30, red to red, holding both housings (see the diagram on the next page).
- 4 Push the junction and the folded pigtail into the bay ahead of the pack, then the pack, ribbon last. The XT30 pair ends up flat on the pack top under the run button, pigtail (female) end to the rear, fuse sleeve beside it, the pigtail fold above. The pigtail must leave the XT30 female straight (no bend at its solder cups): if it does not, stop (gate G-MP-PACK).
- 5 Look up the mouth: nothing may hang in the gaps beside the pack.
- 6 Slide the cap on rearward until the detent clicks.

CAUTION

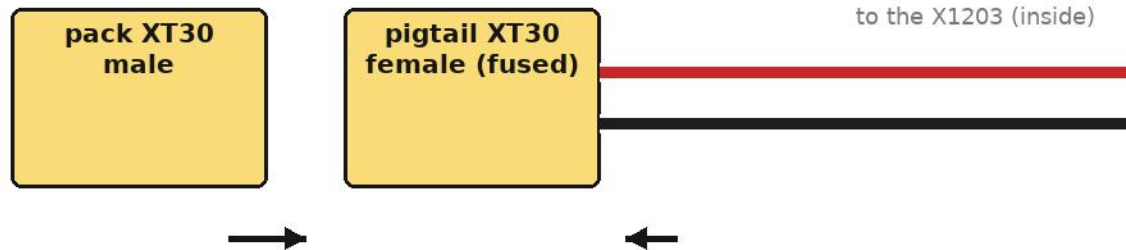
This powers the camera: the X1203 may start the Pi at once.

CHECK

- ☐ whole XT30 housing + 15 mm of wire out before mating
- ☐ no loop beside the pack
- ☐ cap detent clicked

10b XT30 junction

Step 10



Red to red. Push the housings together at the grip mouth; never push or pull by the wires. This connection powers the camera: the X1203 may start the Pi at once.

DO THIS

- 1 Mate by the housings, red to red.

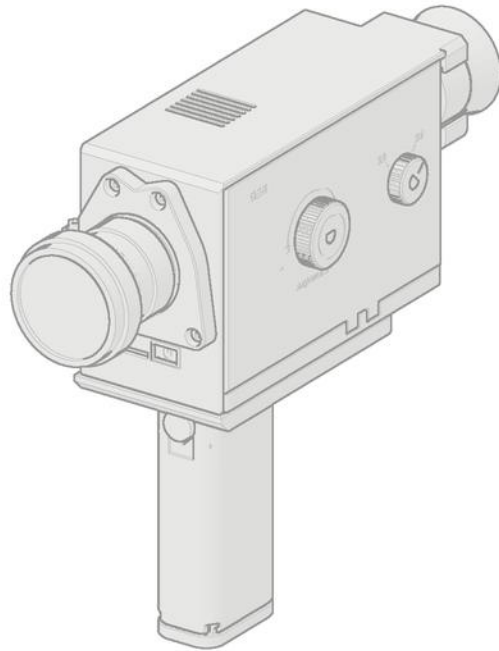
CHECK

- ☐ fully mated, no wire strain

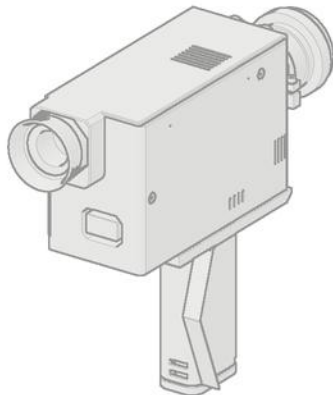
10c Power on, level check, gates

Step 10

A from the front-left, above



B from the rear-right, above



C from above



YOU NEED

- Tools: (1) PH1 screwdriver, (13) Adjustable torque screwdriver 0.1-0.6 N m
- Torque: s_c4 0.2 N m

DO THIS

- 1 Press the plunger: the EVF shows the boot screen (there are no LED indicators).
- 2 Level the horizon on live view every time: loosen s_c4 half a turn, turn lens and camera together until it is level (window ± 1.4 deg; a soft stop at about 3 deg means the tab is on a tine: come back; the lock-screw heads keep 0.3 mm or more off the tab-catch tines), push the lens back onto the cone, re-snug s_c4 to 0.2 N m.
- 3 Run WIRING gates G-W8 to G-W12 in the closed body (G-W12: full-workload power test).

CAUTION

Shut down before service: hold the plunger 2 s, wait for the EVF to go dark, wait 5 s more, then cap off, pack out, XT30 parted (ASSEMBLY s7 P1-P4).

Set the camera down on its right side; never on its left side (the exposure knob), and never stand it on its grip end (it tips at about 6 deg).

CHECK

- ☐ boot screen on the EVF
- ☐ horizon level
- ☐ G-W8 to G-W12 recorded



Before any service

ASSEMBLY s7: isolation first, every time (P1-P4)

P1 Shut down

Hold the plunger 2 s. The EVF shows the shutdown screen, then goes dark. Wait 5 s more. If the EVF stays lit after the shutdown screen went away, do not press again (a press while halted boots the Pi): wait 5 s and go on.

P2 Cap off

Slide the cap forward 52 mm off the grip.

P3 Pack out

Pull the pack out of the bay by its ribbon; the XT30 junction follows it to the grip mouth.

P4 XT30 apart

Unplug by the housings, never by the wires. The pack stays out until the body is closed.

EVF out (s7 item 7)

Unplug the HDMI in place first: smooth-jaw long-nose pliers (tool 15) on the plug's deep half, pull straight down while a fingertip from the front backs the slab. Then slide the OLED and board out together; with the plug on, the pair cannot pass the rail.

Hood release (BX-17)

If a release pin does not stay in its hole, hold both pins with one strip of tape across the right wall, then lift the hood straight up.

Then

Service is the exact reverse of the build (ASSEMBLY s7 items 2-11). Lens swap without isolation: loosen s_c4 one turn, tip the body nose-down, carry the lens, unscrew with fingertips only (the hood tab catch holds the camera through its metal lock tab); a stuck thread: stop, never more than fingertip torque. Refit: lens seated, turn lens and camera to the middle of their free roll, then s_c4.



Physical blocker check

Review of this build sequence, 2026-10-08 (full text: [cad/g8-d2-v1/guide/BLOCKERS-2026-10-08.md](#))

Review verdict (2026-10-08): 1 BLOCKER, 4 MAJOR, 12 MINOR, 16 CLEARED. The D2 can be assembled, but not exactly as written. The one blocker is step 6: the HDMI plug cannot be mated outside the body and then slid in with the EVF pair. The mated plug hits the tub's EVF bottom rail about 24.5 mm before home. The `evf_pair_in` sweep missed this because the plug is not in its moving set (BX-1). The same clash blocks service item 7 in reverse. The four MAJOR items each need a sequence or hardware change in the guide. (BX-2) The 150 mm QT lead is too short to plug the encoder with the panel held "beside the body". (BX-3) Mating the XT30 at the grip mouth is not length-checked, and the step-1 tie "to a standoff" can leave the junction about 28 mm inside the bay. (BX-4) The step-8 hold "by the metal lens mount" has no clean grip: the 10 mm housing band sits between the front wall and a PCB/cover that stand proud of it, with about 6 mm free below it. (BX-5) Pressing `knob_exp` on at step 9 loads only the encoder's 0.4 mm cradle hooks, with nothing behind them. The straight-driver rule holds: all 11 screws (including `s_c1..s_c4`, `s_j`, `s_k1` and `s_k2`) are computed at their own step in the tool-13 envelope (C-1). No solder joint is needed after the body is closed. Everything here is computed or estimated from CAD proxies: nothing was printed or bought, and no hand ever entered the model.

Status after r7 (2026-10-09)

r7: BX-1 fixed in CAD (SPEC-C1) — no geometry change: the EVF pair slides in with the HDMI unplugged, then the HDMI is plugged up through the rail gap in the body; `evf_pair_in` passes, the planted `r6` order fails (60-66 mm3 of tub), new `mate_paths` check: plug path 0 mm3, every hand/tool corridor 0 mm3, 10.0 mm fingertip room; fix-up: no floor-edge lever (the tub top at z 36 is only the stick-guide block), the push comes from the guide top or straight up from the open well (`acc_hdmi_evf_push_well`, 43 mm room; tool 10 only as an upright push stick), service pliers corridor `acc_hdmi_evf_pliers_svc`; `mate_paths` 13/13 in the trial build; first-print gate G-EVF-3.

r7: BX-7 fixed in CAD (SPEC-C1) — the foam pad is stuck to the OLED back (PSA, BOM D2-45) and moves with the pair in `evf_pair_in` and `evf_out`.

r7: BX-9 fixed in CAD (SPEC-C1) — the 5 V PH junction is mated with the pair held about 60 mm out, rides in above the bottom rail and is pushed into `ko_5v_end` with the 120 mm tweezers once the board is home (`acc_5v_tuck` corridor 0 mm3); its reach row `usb_5v_evf` comes with MATE_POSES (SPEC-C2 slot).

r7: C-16 fixed in CAD (SPEC-C1) — `s7` item 7 now unplugs the HDMI in place first (pliers, tool 15, fingertip backing the slab), then slides the pair out, then the eyepiece; removals `evf_out` passes with the eyepiece as an obstacle.

r7: BX-5 fixed in CAD (SPEC-C5) — both knobs are pressed on at bench step 2 with a thumb behind the encoder board, on a 0.2 mm paper shim, and ride in with the panel (`panel_on` sweep 0 mm3); new handling check: `knob_exp` press 0.0 mm3 on rigid parts (1.44 on the snap-tooth zones, thumb column 0.0), `knob_fps` rigid 15.92; physical gate G-ENC-1 pull-out ≥ 25 N.

r7: BX-13 fixed in CAD (SPEC-C5) — the panel screws are driven with the camera on its hood roof on a folded cloth, a block ≤ 25 mm against the left side, stopped from sliding; fix-up: at the bench edge, right face within about 20 mm of it, so the `s_r1/s_r2` driver handle and hand pass beyond the edge (handling driver-plane rule, overhang -Y 20 mm; `hood_down` margin 2.7, tip angle 29.2 deg); set down only on its right side (`right_down` 8 52.6 deg, `right_down_10` 39.8 deg); never on its left side, never on its grip end (it tips at about 6 deg, so base-down is hand-held); driver `s_b1/s_b2/s_r1/s_r2` pass.

r7: BX-14 fixed in CAD (SPEC-C5) — the 18/24 shaft is measured in place (P), then cut off the panel in soft vise jaws (P - 6.9) and the switch refitted; final 6.7-7.1 above the face, recorded at G-MP-SW.

r7: BX-2 fixed in CAD (SPEC-C2) — the QT lead is 250 mm (240-270, Adafruit 4397 spliced to Pololu 5521, WIRING W-QT) and is plugged with the panel 60 mm off; fix-up: a helper holds the body upright, the panel is propped on a block about 110 mm tall, the PH junction is mated first with both hands (`fps_ph` two pinch boxes), then the QT plug with a fingernail push; `mate_reach qt_enc` slack 35.2 at 240 mm, hand box 0 mm3 and wholly outside the body; spares fold into `ko_lead_stow` (`cable_stow` 2288 $\leq$ 2542 mm3); sweeps `panel_on` carries the plug and passes. Physical gate G-QT-1.

r7: BX-10 fixed in CAD (SPEC-C2) — single 1-pin header housings (a 1x2 only on 33/34 or 37/39), seated with open tweezer tips straddling the wire and tug-tested at step 4; `header_housings` gaps cooler 0.68, kit 0.75, wall 1.13, hood 2.27 (≥ 0.3), plus completeness rows (each used pin exactly one housing; 10 rows pass); `ko_run_leads` widened to x -64.5. Physical gate G-HDR-1.

r7: BX-3 fixed in CAD (SPEC-C3) — the pigtail is cut to `L_cut` = 200 mm + the G-W2 pads-to-W length and strain-relieved by a neutral-cure RTV bead at the pads (no standoff tie); new `lead_access` check: XT30 face 28.4 mm out of the mouth (≥ 15), required length 190, cut range worst 17.4; fix-up: `pigtail_drop` stops the cut above 25 mm pads-to-W (G-W5 drop 0.15 V), the junction is re-posed flat on the pack top under the run button, female end to the rear, fuse beside it (`pigtail_store` fill 0.149, exits 8.0 / 8.0 mm against 7.2), step 10 is done on the right side with the pack in the palm; `mate_reach xt30` slack 13.4. Physical gates G-W2, G-MP-PACK (rigid XT30 exit 0.8 mm at most: to measure).

r7: BX-11 fixed in CAD (SPEC-C3) — the pigtail leg is formed into a U at the W mark (x -37.5) and taped under the X1203 at bench step 1, so it comes down with the stack; the lane moved 0.5 off the keeper (`pigtail_wrap_clear` 0.000, `lane_separation` 1.0); `pi_in` carries the leg as riders, 0 mm3.

r7: BX-16 fixed in CAD (SPEC-C3) — the run lead is threaded up through the base passage and the rear end of the floor hole (now x -41.5..-28) before the tub is lowered, then taped in its lane; `run_lead_window` 3.5 $\geq$ 3.2 at the start of the slide; `base_grip` STL unchanged.

r7: BX-4 fixed in CAD (SPEC-C4) — the hood roll fin is replaced by a two-tine tab catch, so the camera's metal lock tab takes the lens-thread torque and nobody holds the camera; new `roll_catch` check: first contact 3.25 deg (1.76..4.77), cover/PCB never first, window 0.679 / 0.129; fix-up: step 8 turns lens and camera back off the tine before `s_c4` (`lens_turn_back_lint`), the step-10 level check is done every time, `lens_hold_lint` also scans the guide, and `roll_catch` is in the audit's expected categories; physical gates G-CAM-1 (head side), G-CAM-2.

r7: not fixed in CAD this round: BX-6, BX-12, BX-17 (SPEC-C6 not landed), BX-15 (SPEC-C4 5b: its part-done hunks were rolled back at the release, so the collar gauge stays the `r6` part) and BX-8 (SPEC-C4 5c, not landed). Their interim tips stay on guide pages 3b, 5a, 7a, 7b and B0.

Original findings (2026-10-08)

! Physical blocker check (continued)

BX-1 [BLOCKER] step 6 (and s7 item 7): The EVF board cannot slide home with the HDMI plug already mated "outside the body". The mated right-angle plug hangs below the board's lower edge (ko_hdmi_evf: x -144.8..-136.8, z 52..63.8). On the -Y 45 slide it meets the tub's EVF bottom rail. The rail is gapped only at y 3..18 (gap 4..17 + 1.0 clear), and its body below the groove floor stays full width from y 18 to its end at y 28.5. The plug stops about 24.5 mm before home. -> CAD: extend the bottom-rail cut under the plug's x span (x <= px1) over the full rail height from g1 to yend (keep the +X spine), and add the mated plug (ko_hdmi_evf box) to the evf_pair_in moving set. Guide, until fixed: mate only the OLED flex outside. Slide the pair home, then plug the HDMI upward into the board through the rail gap (y 4..17) from the open left side; check by hand that this works on the first print.

BX-2 [MAJOR] step 8: QT lead too short for "hold the panel beside the body" while plugging the JST-SH into the encoder. The lead is dressed at step 4 along the right wall and across behind the blower inlet. The estimate stops at the encoder envelope (z 44), not at the socket. The modelled QT sockets are on the encoder back at mid-height (z 54..60), about 14-17 mm further along the lead. So the free end reaches only about 0-7 mm past its final socket position, or about 15-22 mm if the 15 mm end allowance is used up. The socket sits about 8-12 mm inside the panel's inner face. The panel can therefore be held only about 15-20 mm off its seat: a slot too narrow for fingers to push a 1.0 mm-pitch JST-SH home. Nothing computes this pose. -> Best: a 200-300 mm JST-SH-to-socket lead (re-run cable_routes with slack >= 40 mm), coiled in ko_lead_cross. Guide, with the 150 mm lead: before step 5, leave the cross run loose (not tied), pull the QT end out, tilt the panel out from its bottom edge (lip side) about 25 deg rather than holding it off parallel, plug the JST-SH with tweezers backing the socket, then fold the lead back while the panel goes on. Mate the 18/24 PH (slack 47.7 mm) first.

BX-3 [MAJOR] step 1, 10 (s5 Fit, s7 P3-P4): Grip-mouth XT30 mating is not length-checked, and the step-1 "strain-relief tie to a standoff" can make it impossible. Taut, the 180 mm pigtail leaves about 28.6 mm of wire past the grip mouth (no tie detour, assumed pad position). The nearest kit standoffs are 28.7 / 29.7 mm from the route's wrap point at the port edge (x -38). A tie there and back costs about 57 mm, which leaves the XT30 about 28 mm inside the bay: a blind mate in a 25 x 43 mm bay, and P4 "unplug by the housings" becomes hard. -> Define the tie point on the route (a tie slot by ko_pig_wrap, or on the pi_keeper), not "a standoff". Or set the pigtail to about 220 mm after G-W2 locates the pads; check that the extra coil fits ko_xt30 above the pack. Add a computed "XT30 face >= 15 mm out of the grip mouth" check. Guide: at G-W2, measure pads-to-mouth with a string before cutting the pigtail.

BX-4 [MAJOR] step 8 (also s7 6a, lens swap): "Finger and thumb hold the camera by its metal lens mount" cannot be done as written. The reachable metal is a band 10.35 mm long (housing dia 35.5, x -17.2..-6.85). It sits between the front wall (x -5.2) and the PCB/cover, which stand 1.25 / 2.0 mm proud of it on every side (PCB 38 sq, cover 39.5 sq). A fingertip (about 14-16 mm) will also bear on the PCB/cover edge. Below the band there is about 6 mm to the cooler/exhaust (z 42.25 vs 36.3), so no thumb fits under it. The right side is reachable only by curling a finger over the top (about 19.5 mm gap to the hood roof). So the B-7 promise (no thread torque through the housing-to-PCB joint) is not physically assured. -> Guide: thumb on the left side of the mount band, index finger over the top and down the right side of the band. Press forward (+X) only; never squeeze the cover. Thread the lens with fingertip torque only until it stops, with no final "snug". Better (CAD): a printed "camera steady" fork for the 10 mm band, inserted from the left, or a grip flat on the BFAR.

BX-5 [MAJOR] step 9: Pressing knob_exp onto the encoder shaft with the panel closed loads only the 3 cradle hook teeth (0.4 mm). The press pushes the encoder inward (-Y), away from the panel face. Nothing inside the body backs the encoder: the HDMI run below it is a keep-out, and there is no hook under the PCB. A firm press fit can pop the encoder off its cradle inside the closed body; recovery means panel off (4 PT drives on the tally). -> Guide: press knob_exp on at step 2 (bench, panel off), backing the encoder PCB with a thumb. The knobs then travel with the panel (the panel_on path stays outside the body) and "ride out on their shafts" in service (s7 item 4). knob_fps on the nut-held switch is not affected. If it stays at step 9: twist the knob on lightly; never push hard.

BX-6 [MINOR] step 5: The plunger has no retention while the hood plate comes down. The flange sits in a 0.4 mm recess, and the finger holding it must leave before the plate's channel reaches it, so the plunger can fall out forward. -> Guide: tilt the body nose-up (rest it on the rear face) so the plunger stays in by gravity, then lower the hood straight down relative to the body.

BX-7 [MINOR] step 6: Foam pad order is ambiguous. The pad (1.0 mm) shares the 3.0 mm OLED pocket with the OLED (1.93 mm), but the sweeps treat it as already placed (an obstacle for eyepiece_in and evf_pair_in). A loose pad has nothing to stand against before the board arrives, and a sliding OLED would drag it. -> Guide: stick the pad to the OLED back (or the board's -X face) first, then slide it in as part of the tethered pair.

BX-8 [MINOR] step 7: The 9-layer FPC S-fold (about 119 mm of slack) has to be formed in ko_fpc_loop (12.8 x 17.5 x 19.4 mm) through the open left side, under the hood, behind the camera. -> Guide: pre-fold the S-fold on the bench (bend r >= 1, hold it with a Kapton strip) before the camera goes in; check that it does not touch the blower inlet.

BX-9 [MINOR] step 6: Tucking the 5 V PH junction into ko_5v_end (x -136.5..-130, y -24..1.5, z 64..70) means reaching 30-56 mm in from the left opening, under the hood, just ahead of the board that has just gone in. -> Guide: use the 120 mm tweezers. Lay the junction before the board's last 10 mm of travel.

BX-10 [MINOR] step 4: Header sockets are tight. The even-row Dupont housings (pin 6, 34) sit about 1.2 mm from the right-wall inner face (y -32.5), and the odd row (1/3/5, 33/37/39) about 0.7 mm from the cooler edge (y -25.55). Fingers cannot grip the housings from the sides. -> Guide: seat each housing straight down with closed tweezer tips on its top, then pull-test each one. Single 1-pin housings (or a 1x2 for 37/39) are fine; do not use a 1x3 or 2x3 shell.

BX-11 [MINOR] step 4: Hold the stack by its USB end and port edge only: its right and front edges end 0.2 / 0.35 mm from the walls. The pigtail and run lead must lie flat under the X1203 (ko_pig_under / ko_run_floor, 3.1 mm tall) while the stack comes down the last 5 mm, and that space is blind. -> Guide: pre-form both leads and hold them to the floor with Kapton tape. Pull the XT30 gently down from the grip mouth as the stack descends. Check 4: look under the port edge before the keeper goes in.

! Physical blocker check (continued)

BX-12 [MINOR] step 3, 8: Driver-handle margins depend on the real torque screwdriver. The s_j handle is 2.35 mm from the strap (flexible); the s_c4 handle is 6.5 mm from the hood. -> Buy-check tool 13: handle dia ≤ 30 . With a 1/4 in hex holder, use a ≥ 50 mm PH1 power bit whose turned shank (≤ 6.5) covers at least the first 8 mm; the counterbores are dia 7, up to 6 mm deep. Hold the strap aside for s_j.

BX-13 [MINOR] step 8: Before the knobs are on, the encoder D shaft (to y 44) and the cut switch shaft (to y 42) stand 7-9 mm proud of the panel. Laying the body on its panel side to drive s_r1/s_r2 vertically rests it on the shafts. The eyepiece is also loose rearward until the panel's EVF cap clamps it. -> Guide: drive s_r1/s_r2 with the body upright (driver horizontal), or on a folded towel with a gap for the shafts. Keep the body level or nose-down between steps 6 and 8.

BX-14 [MINOR] step 2: Sawing the 18/24 shaft after the switch is fitted puts the hacksaw load into the nut, the anti-rotation tab and the printed panel. -> Guide: support the panel face on a block next to the shaft, or cut the shaft in a vise before fitting (7.0 above the panel face; measure the nut/washer stack).

BX-15 [MINOR] step 7: Thumb on the centring gauge vs the collar-screw lines: the gauge is about 44.3 dia x 24.4 long (front face about x 19.8). The s_c1/s_c2 bits pass 8 mm above its rim. The s_c3 line (y 28, z 36) passes the lower-left of the gauge face. -> Guide: thumb on the upper-right of the gauge face while driving s_c3; on the lower half while driving s_c1/s_c2.

BX-16 [MINOR] step 3: The run lead (250 mm, two Dupont sockets) has to stay clear of the tub-floor / base interface during the 10 mm slide, and is then fished blind through a 6 x 7.5 mm hole. -> Guide: before the slide, let the socket end hang down the grip bay (not lying flat on the base). After s_j, push it up from the open grip mouth with a finger while the tweezers take it through the floor hole.

BX-17 [MINOR] step s7 item 9 (service): Hood release: the s7 text says the dia 1.5 pins "stay in by friction" in dia 1.6 holes while the hood is lifted. The deflected hook tooth pushes each pin back out along its axis, and the nominal 0.1 mm clearance gives no designed friction, so the two releases may not stay made. Then one person needs both pins held plus the hood lifted. -> Guide: if a pin does not stay in, hold both pins with one strip of tape across the right wall (or a rubber band round the body), then lift the hood. Physical gate G-SNAP-2 should record whether the pins stay.